

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2021 - 2022)

JEE(Advanced)

SCORE-III

21-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	C	D	B,D	B,C	A,C	B,C	A,B,C	A,B,D
SECTION-II	Q.	11	12								
	A.	A,C	A,C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	3.07 to 3.08	20.50	45.00	-4.00	9.00	2.00				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4					
	A.	A	C	D	C					
	Q.	5	6	7	8	9	10	11	12	
	A.	B,D	A,C,D	A,B,D	B,C,D	A,B,D	A,B	A,C,D	A,C,D	
SECTION-II	Q.	1	2	3	4	5	6			
	A.	8.00	400.00	43.00	3.00	7.00	8.00			

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4					
	A.	B	D	B	A					
	Q.	5	6	7	8	9	10	11	12	
	A.	A,B,C,D	A,C	A,C,D	A,B,C,D	A,C,D	C,D	B,D	A,C,D	
SECTION-II	Q.	1	2	3	4	5	6			
	A.	87.00	144.00	150.00	5.00	4.00	0.50			

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Test Pattern



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JEE(Main+Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,D	B,D	B,C	A,D	A,C	A,C	A,D	B,D	B	C
SECTION-II	Q.	11	12								
	A.	C	A								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	8.00	3.00	40.00	12.00	3.12	28.28				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	
	A.	A,B,D	B,D	B,C	A,C,D	B,C	A,B,C	B,C,D	C,D	
	Q.	9	10	11	12					
	A.	B	D	C	B					
SECTION-II	Q.	1	2	3	4	5	6			
	A.	100.00	8.00	2.00	375.00	4.00	19.00			

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B	A,C	A,B,C,D	A,D	A,B,C	A,B,C	A,C,D	A,B,C	D	C
SECTION-II	Q.	11	12								
	A.	D	C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	1.00	16.00	16.50	3.00	6.00	3.00				

	CLASSROOM CONTACT PROGRAMME (Academic Session : 2020 - 2021)	JEE(Advanced) FULL SYLLABUS 21-08-2021
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JEE(Main+Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH
PAPER-1
PART-1 : PHYSICS
SOLUTION
SECTION-I
1. Ans. (C)

Sol. $\frac{1}{\lambda} = R \left(1 - \frac{1}{2^2} \right) = \frac{3R}{4}$

As per Doppler's effect, $\frac{v \cos \theta}{c} = \frac{\Delta \lambda}{\lambda}$

$$\therefore v = \frac{c \Delta \lambda}{\lambda \cos \theta} = \frac{3CR \Delta \lambda}{4 \cos \theta}$$

$$= 6.98 \times 10^5 \text{ m/s}$$

2. Ans. (A)

Sol. EF at distant $r = \frac{\rho r}{2\epsilon_0} = \frac{(n_0 e)r}{2\epsilon_0}$

and MF at distance r can be obtained by

$$B(2\pi r) = \mu_0(\pi r^2(en_0)v)$$

$$\therefore B = \frac{\mu_0 r e v n_0}{2}$$

Net force on (e^-) = electrostatic force – Magnetic force

$$= \frac{e^2 n_0 r}{2\epsilon_0} \left(1 - \frac{v^2}{c^2} \right) \left[\frac{1}{\mu_0 \epsilon_0} = c \right]$$

$$\text{so, } \frac{n e^2 r}{2\epsilon_0} = \frac{e^2 n_0 r}{2\epsilon_0} \left(1 - \frac{v^2}{c^2} \right) \Rightarrow n = n_0 \left[1 - \frac{v^2}{c^2} \right]$$

3. Ans. (C)

Sol. Net charge will remain same

Suppose initial energy is given by

$$U_0 = \frac{kq^2}{a} \times C \quad C = \text{dimensionless constant}$$

New energy

$$U = \frac{kq^2}{a/\sqrt{2}} \times C = \sqrt{2}U_0$$

4. Ans. (D)

Sol. The magnetic field due to the magnet at a distance x in the broad side on position is given by

$$B = -\frac{\mu_0}{4\pi} \cdot \frac{M}{x^3}; \text{ here } M \text{ is opposite to the field (see figure).}$$

Consider a shell of radius x and $x + dx$

Area of the shell = $2\pi x dx$

$$\text{Magnetic flux through the sell } \Delta \phi = -\frac{\mu_0}{4\pi} \cdot \frac{M}{x^3} \cdot 2\pi x dx$$

$$\text{Net flux through the coil} = \frac{\mu_0}{2} M n \int_{r_0}^r \frac{dx}{x^2} = \frac{\mu_0}{2} M n \left[\frac{1}{r_0} - \frac{1}{r} \right] = \frac{\mu_0 M n (r - r_0)}{2 r r_0} \cong \frac{\mu_0 M n}{2 r_0}$$

On removing the magnet the final flux is zero

$$\therefore \text{Change in flux } \phi = \frac{\mu_0}{2} \cdot \frac{M n}{r_0}$$

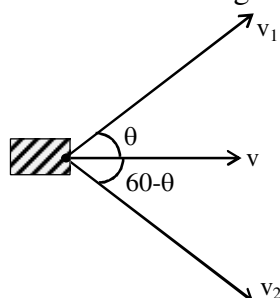
$$q = \frac{\phi}{R} = \frac{\mu_0}{2} \cdot \frac{M n}{r_0 R} \Rightarrow M = \frac{2 q r_0 R}{\mu_0 n}$$

5. **Ans. (B, D)**

6. **Ans. (B, C)**

7. **Ans. (A, C)**

Sol. Let the velocity of the block be v in a direction making an angle θ with the direction v_1 .



For a string to be taut, the objects at the two ends must have same velocity component along the length of the string

$$\therefore v \cos \theta = v_1 \quad \dots\dots\dots(1)$$

$$\therefore v \cos (60 - \theta) = v_2$$

$$\Rightarrow v \left[\frac{1}{2} \cos \theta + \frac{\sqrt{3}}{2} \sin \theta \right] = v_2$$

$$\text{using (1) } \cos \theta = \frac{v_1}{v} \text{ and } \sin \theta = \frac{\sqrt{v^2 - v_1^2}}{v}$$

$$\therefore v \left[\frac{1}{2} \frac{v_1}{v} + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{v^2 - v_1^2}}{v} \right] = v_2$$

$$v_1 + \sqrt{3} \sqrt{v^2 - v_1^2} = 2v_2$$

$$3(v^2 - v_1^2) = 4v_2^2 + v_1^2 - 4v_1v_2$$

$$\Rightarrow 3v^2 - 4v_2^2 + 4v_1^2 - 4v_1v_2$$

$$\therefore v = \frac{2}{\sqrt{3}} \sqrt{v_1^2 + v_2^2 - v_1v_2} \quad \dots\dots\dots(2)$$

(b) For $\theta = 0^\circ$

$$v = v_1 \quad [\text{from (1)}]$$

Putting into (2) we get

$$v_1 = \frac{2}{\sqrt{3}} \sqrt{v_1^2 + v_2^2 - v_1v_2}$$

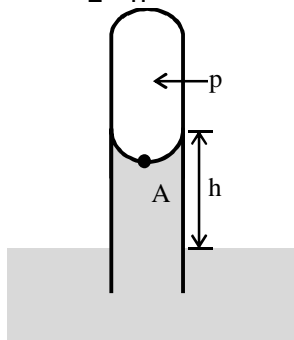
$$\frac{3v_1^2}{4} = v_1^2 + v_2^2 - v_1v_2 \Rightarrow v_1^2 - 4v_1v_2 + 4v_2^2 = 0 \Rightarrow (v_1 - 2v_2)^2 = 0$$

$$\Rightarrow v_1 = 2v_2 \Rightarrow \frac{v_1}{v_2} = 2$$

8. Ans. (B,C)

Sol. When the tube is brought into contact with water, it is filled with air at atmospheric pressure. When water rises to a height h , the air pressure (P) is given by

$$\therefore P = \frac{P_0 L}{L-h}$$



Radius of curvature of meniscus $R = r$, since contact angle is zero.

Pressure at A is $P_A = P_0 - \rho gh$

$$\therefore P = P_A + \frac{2T}{r}$$

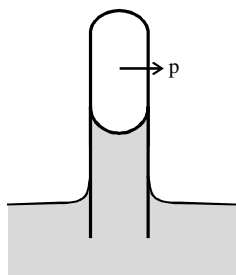
$$\therefore \frac{P_0 L}{L-h} = P_0 - \rho gh + \frac{2T}{r} \Rightarrow P_0 \left(\frac{L}{L-h} - 1 \right) = \frac{2T}{r} - \rho gh$$

$$\Rightarrow \frac{P_0 h}{L-h} = \frac{2T}{r} - \rho gh \Rightarrow L-h = \frac{P_0 h}{\frac{2T}{r} - \rho gh}$$

$$\Rightarrow L = h + \frac{P_0 r h}{2T - \rho gh}$$

Surface tension force on outer wall of the tube is $2\pi rT(\downarrow)$

Surface tension force (at meniscus) on the inner tube wall is $2\pi rT(\downarrow)$



Force due to air pressure inside the tube $\pi r^2 P(\uparrow)$

Force due to air pressure inside the tube $\pi R^2 P_0(\downarrow)$

For equilibrium of the tube, let the upward force needed be F

$$F + P\pi r^2 = Mg + \pi R^2 P_0 + 2\pi rT + 2\pi rT$$

$$F = Mg + \pi R^2 P_0 - \frac{P_0 L \pi r^2}{L-h} + 2\pi(R+r)T$$

$$F = Mg + \pi P_0 \left[R^2 - \frac{Lr^2}{L-h} \right] + 2\pi(R+r)T$$

9. Ans. (A, B, C)

Sol. $R_{\max} = a + ae$

$$R_{\min} = a - ae$$

$$a = \frac{R_{\max} + R_{\min}}{2} ; b = \sqrt{R_{\max} \cdot R_{\min}}$$

$$\therefore v_{\max} = \sqrt{GM \left(\frac{2}{R_{\min}} - \frac{1}{a} \right)} = \sqrt{\frac{GM(2a - R_{\min})}{aR_{\min}}} = \sqrt{\frac{GMR_{\max}}{aR_{\min}}} \quad \dots(1)$$

$$\text{as } R_{\max} + R_{\min} = 2a$$

$$v_{\min} = \sqrt{GM \left(\frac{2}{R_{\max}} - \frac{1}{a} \right)} = \sqrt{\frac{GM(2a - R_{\max})}{aR_{\max}}} = \sqrt{\frac{GMR_{\min}}{aR_{\max}}} \quad \dots(2)$$

Multiplying (1) and (2)

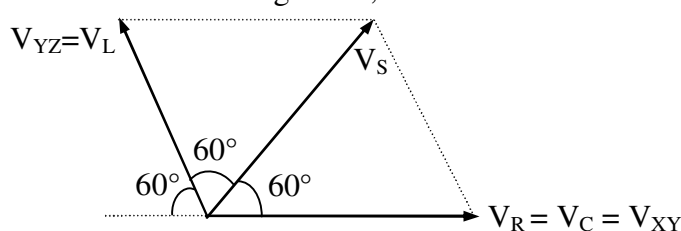
$$\text{or } \sqrt{v_{\max} v_{\min}} = \sqrt{\frac{GM}{a}} = a \sqrt{\frac{GM}{a^3}} = \frac{2\pi a}{T}$$

$$\text{as } T = 2\pi \sqrt{\frac{a^3}{GM}} \quad \therefore a = \frac{T}{2\pi} \sqrt{v_{\max} v_{\min}}$$

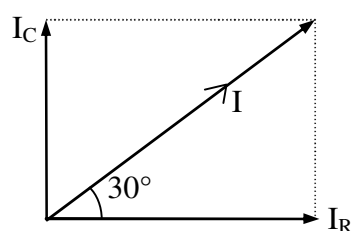
10. **Ans. (A, B, D)**

Sol. $\vec{V}_{\text{source}} = \vec{V}_{XY} + \vec{V}_{YZ}$

Since all has same magnitude,



and since $\vec{V}_L$ is perpendicular to $\vec{I}$,



so ϕ between I and V_s is 30°

$$\therefore \langle P \rangle = V_s I \cos 30^\circ$$

$$= 100(5) \frac{\sqrt{3}}{2}$$

$$= 250\sqrt{3} \text{ watt}$$

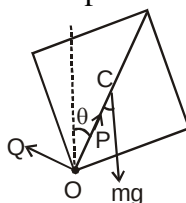
$$X_C = \frac{V_C}{I_C} = \frac{100}{I \sin 30^\circ} = \frac{100}{5/2} = 40\Omega$$

$$I_R = I_{R_0} \sin(\omega t - 60^\circ)$$

$$= I_0 \cos 30^\circ \sin\left(\omega t - \frac{\pi}{3}\right) = \frac{5\sqrt{3}}{\sqrt{2}} \sin\left(\omega t - \frac{\pi}{3}\right)$$

11. **Ans. (A, C)**

Sol. Let P and Q be the components of reaction of pivot on the cube along and perpendicular to the OC.



M.I. of the cube about O is $\frac{2}{3}Ma^2$

$$\alpha = \frac{\tau}{I} = Mg \sin \theta \cdot \frac{a}{\sqrt{2}} / \frac{2}{3}Ma^2 = \frac{3}{2\sqrt{2}} \frac{g \sin \theta}{a}$$

Now along OC, $Mg \cos \theta - P = M \frac{a}{\sqrt{2}} \omega^2$

Perpendicular to OC, $Mg \sin \theta - Q = M \frac{a}{\sqrt{2}} \alpha$

$$Q = (Mg \sin \theta)/4$$

Integrating α or using energy conservation, we get $\omega^2 = \frac{3g(1 - \cos \theta)}{a\sqrt{2}}$

$$P = Mg \cos \theta - Ma \frac{3g(1 - \cos \theta)}{2a} = \frac{-3}{2}Mg + \frac{5}{2}Mg \cos \theta$$

When $P = 0$ $\cos \theta = 3/5$, $\sin \theta = 4/5$

Now horizontal component of reaction is $P \sin \theta - Q \cos \theta = (-3/25)Mg$

Vertical component of reaction is $P \cos \theta + Q \sin \theta = (4/25)Mg$

12. **Ans. (A, C)**

Sol. $\beta = \frac{dP}{\left(\frac{d\rho}{\rho}\right)}$

$$\therefore \int_{\rho_0}^{\rho} \frac{d\rho}{\rho^2} = \frac{\int g dy}{\beta}$$

$$\therefore -\frac{1}{\rho} \Big|_{\rho_0}^{\rho} = \frac{gy}{\beta}$$

$$\therefore \frac{1}{\rho_0} - \frac{1}{\rho} = \frac{gy}{\beta}$$

$$\therefore \frac{1}{\rho_0} - \frac{gy}{\beta} = \frac{1}{\rho} \quad \therefore \rho = \frac{\rho_0}{1 - \frac{\rho_0 gy}{\beta}} = \rho_0 \left(1 + \frac{\rho_0 gy}{\beta} \right)$$

Now, $dp = \rho g dh$

$$\int_{p_0}^p dp = \rho_0 g \int \left(1 + \frac{\rho_0 gh}{\beta} \right) dh$$

$$\therefore p = p_0 + \rho_0 hg \left[1 + \frac{\rho_0 gh}{2\beta} \right]$$

SECTION-II

1. **Ans. 3.07 to 3.08**

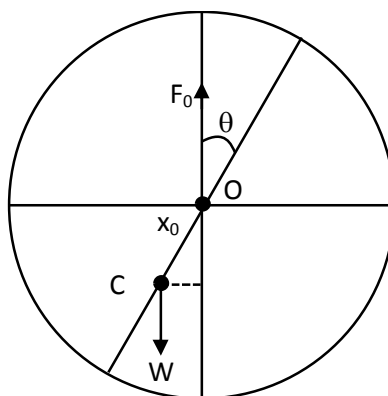
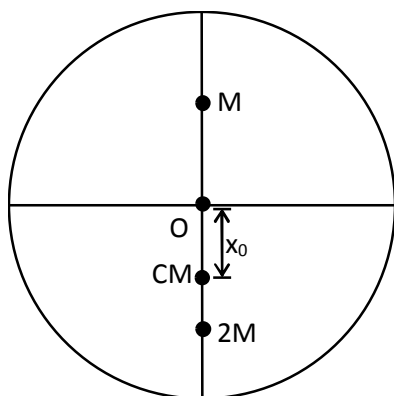
Sol. Let's first find the position of COM of the ball.

$$x_0 = \frac{2M \left(\frac{3R}{8} \right) - M \left(\frac{3R}{8} \right)}{2M + M} = \frac{R}{8}$$

[$\therefore$ distance of COM of each hemisphere = $\frac{3R}{8}$ from point O]

Where M = mass of upper hemisphere

$2M$ = mass of lower hemisphere



Now let's find the moment of inertial(I) of the ball about an axis passing through COM (perpendicular to the fig).

First we will write the moment of inertial (I_0) about an axis through O and perpendicular to the plane of the fig.

$$I_0 = \frac{2}{5}MR^2 + \frac{2}{5}(2M)R^2 = \frac{6}{5}MR^2$$

Using parallel axes theorem

$$I_0 = I + 3M x_0^2 \quad [\because \text{mass of sphere} = 3M]$$

$$\therefore I = \frac{6}{5}MR^2 - 3M \frac{R^2}{64} = \frac{369}{320}MR^2$$

Consider the ball slightly displaced about the equilibrium. The torque due to buoyancy (about the axis through COM) provides the necessary restoring torque.

$$\therefore I \alpha = -F_B x_0 \sin \theta$$

$$\therefore \frac{369}{320}MR^2 \cdot \alpha = -F_B \frac{R}{8} \quad [\text{for small } \theta, \sin \theta = \theta]$$

$$\because F_B = 3Mg \quad \therefore \alpha = -\left(\frac{40g}{123R}\right)\theta$$

$$\omega = \sqrt{\frac{40g}{123R}} \Rightarrow T = 2\pi \sqrt{\frac{123R}{40g}}$$

2. Ans. 20.50

Sol. Zero error (+ve) = $0 + 5 \times 0.005 \text{ mm}$
= 0.025 mm

$$\therefore LC = \frac{\text{Pitch}}{\text{Total LSD}} \text{ LSD} \Rightarrow 0.005 \text{ mm} = \frac{\text{Pitch}}{200}$$

$$\therefore \text{Pitch} = \text{MSD} = 0.005 \text{ mm} \times 200$$

= 1 mm

New observed diameter

$$= (4 \text{ div.} \times \text{MSD}) + (25 \times 0.005 \text{ mm})$$

$$= 4 \text{ mm} + 0.125 \text{ mm}$$

$$= 4.125 \text{ mm}$$

$$\therefore \text{Actual diameter} = \text{observed} - \text{error}$$

$$= 4.125 - (0.025) = 4.10 \text{ mm} = 4100 \mu\text{m}$$

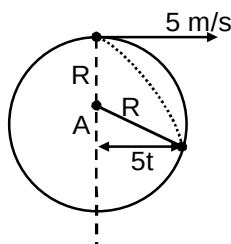
$$= 100 \times$$

$$\therefore x = 41$$

$$\text{Hence ans.} = \frac{x}{2} = 20.50$$

3. Ans. 45.00

Sol.



Using equation of kinematic

$$\frac{1}{2}gt^2 = R + h$$

Using Pythagoras theorem

$$h^2 + (5t)^2 = R^2$$

Using work energy theorem

$$\frac{1}{2}mV^2 - \frac{1}{2}m5^2 = mg(R + h)$$

$$V^2 = 5^2 + 2g(R + h)$$

$$V^2 = 5^2 + 2g\left(\frac{1}{2}gt^2\right)$$

$$V^2 = 25 + 100t^2$$

$$V^2 = 25 + 4[R^2 - h^2]$$

$$V^2 = 25 + 4R^2 - 4h^2$$

$$5t^2 = R + h$$

$$25t^2 = R^2 - h^2$$

$$5 = R - h$$

$$V^2 = 25 + 4R^2 - 4[R-5]^2$$

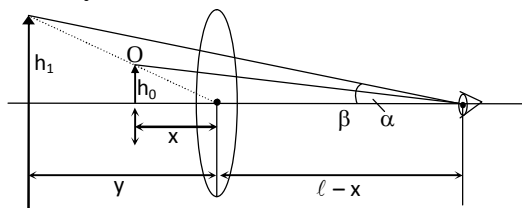
$$25 + 4(3)^2 - 4(3-5)^2$$

$$= 25 + 36 - 4 \times 4$$

$$25 + 20 = 45.00$$

4. Ans. -4.00

Sol. Suppose the distance of object from the lens is x and the distance of corresponding image from the lens is y .



$$\text{Then } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow -\frac{1}{y} + \frac{1}{x} = \frac{1}{f} \Rightarrow y = \frac{fx}{f-x}$$

$$\text{Now } \frac{h_1}{h_0} = \frac{y}{x} = \frac{f}{f-x}$$

$$\text{The angular magnification is } M = \frac{\beta}{\alpha} = \frac{\frac{h_1}{(y+(\ell-x))}}{\frac{h_0}{\ell}} = \left(\frac{h_1}{h_0}\right)\left(\frac{1}{\ell+(y-x)}\right) \Rightarrow m = \frac{f\ell}{x^2 + \ell f - \ell x}$$

From m to be maximum

$$\frac{d}{dx}(x^2 + \ell f - \ell x) = 0 \Rightarrow x = \frac{\ell}{2} \quad \text{Therefore } (m)_{\max} = \frac{1}{1 - \left(\frac{\ell}{4f}\right)}$$

5. **Ans. 9.00**

Sol. By conservation of energy,

$$(m_1 + m_2) g \ell (1 - \cos \theta) = \frac{1}{2} m_1 (5g\ell) \quad \dots\dots (1)$$

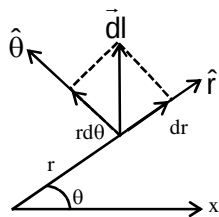
By conservation of momentum,

$$m_1 \sqrt{2g\ell(1 - \cos \theta)} - m_2 \sqrt{2g\ell(1 - \cos \theta)} = -m_1 \sqrt{5g\ell} \quad \dots\dots (2)$$

Solving, $m_2 = 9 \text{ kg}$

6. **Ans. 2.00**

Sol. Unit vectors along $\vec{r}$ and perpendicular to $\vec{r}$ (known as azimuthal direction) are $\hat{r}$ and $\hat{\theta}$ respectively. A small element of length $d\ell$ along the curve can be represented as



$$d\vec{\ell} = (dr)\hat{r} + (r d\theta)\hat{\theta}$$

Field due to such an element at O is

$$d\vec{B} = \frac{\mu_0}{4\pi} I \frac{d\vec{\ell} \times (-\vec{r})}{r^2}$$

$$= \frac{\mu_0 I}{4\pi} \frac{1}{r^2} [(dr)\hat{r} + (r d\theta)\hat{\theta}] \times (-\hat{r}) = \frac{\mu_0 I}{4\pi} \frac{ld\theta}{r} \hat{k}$$

$$B = \frac{\mu_0 I}{4\pi} \int_0^{\pi/2} \frac{d\theta}{b + \frac{c}{\pi} \theta}$$

$$= \frac{\mu_0 I}{4\pi} \frac{\pi}{c} \left[\ell \ln \left(b + \frac{c}{\pi} \theta \right) \right]_0^{\pi/2} = \frac{\mu_0 I}{4c} \left[\ell \ln \left(b + \frac{c}{2} \right) - \ell \ln b \right]$$

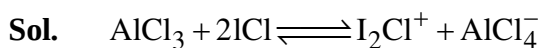
$$B = \frac{\mu_0 I}{4c} \left[\ell \ln \left(1 + \frac{c}{2b} \right) \right] \odot$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. **Ans. (A)**



AlCl_3 produces the solvonium ion in ICl , hence it is an acid in ICl solvent.

2. **Ans. (C)**

3. **Ans. (D)**

Sol. $\text{H}_4\text{P}_2\text{O}_5$ pyrophosphorous acid $\Rightarrow \text{H}_2\text{P}_2\text{O}_5^{2-}$ pyrophosphite ion

(Basicity = 2)

$\text{H}_3\text{P}_2\text{O}_5^-$ Hydrogen pyrophosphite ion

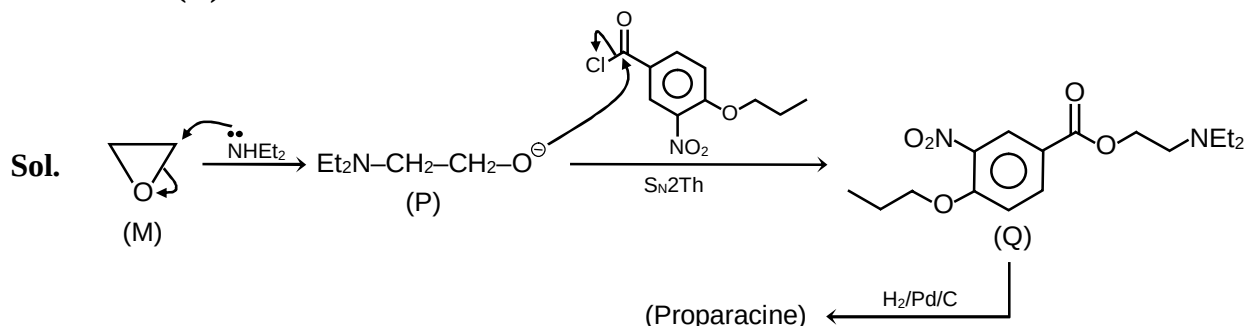
H_3PO_3 phosphorous acid $\Rightarrow \text{H}_2\text{PO}_3^-$ Hydrogen phosphite ion

(Basicity = 2)

$\text{H}_4\text{P}_2\text{O}_7$ pyrophosphoric acid $\Rightarrow \text{H}_3\text{P}_2\text{O}_7^-$ Trihydrogen pyrophosphate ion

(Basicity = 4)

4. Ans. (C)



5. Ans. (B,D)

Sol.	He: $1s^2$	I excited state $\Rightarrow 2s$	Degeneracy = 1
	He ⁺ : $1s^1$	I excited state $\Rightarrow 2 (2s = 2p)$	Degeneracy = 4
	Li ⁺ : $1s^2$	III excited state $\Rightarrow 3s$	Degeneracy = 1
	Li ²⁺ : $1s^1$	II excited state $\Rightarrow 3 (3s = 3p = 3d)$	Degeneracy = 9

6. Ans. (A,C,D)

Sol.	$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	$\Delta S_{\text{system}} \Rightarrow -ve$
		$\Delta S_{\text{surrounding}} \Rightarrow +ve$
	$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$	$\Delta S_{\text{system}} \Rightarrow +ve$
		$\Delta S_{\text{surrounding}} \Rightarrow -ve$
	$H_2O(l) \rightleftharpoons H_2O(s)$	$\Delta S_{\text{system}} \Rightarrow -ve$
		$\Delta S_{\text{surrounding}} \Rightarrow +ve$
	$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	$\Delta S_{\text{system}} \Rightarrow -ve$
		$\Delta S_{\text{surrounding}} \Rightarrow +ve$

7. Ans. (A,B,D)

Sol. Finally (after a very long time) all the solutions will become isotonic

$$\frac{10}{V_A} = \frac{10 \times 2}{V_B} = \frac{10 \times 3}{V_C} = \frac{10 \times 4}{V_D}$$

$$V_A : V_B : V_C : V_D = 1 : 2 : 3 : 4$$

Total volume = 400 ml

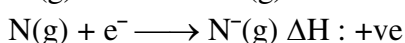
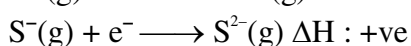
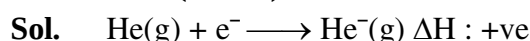
$$V_A = 40 \text{ ml} \quad V_B = 80 \text{ ml}$$

$$V_C = 120 \text{ ml} \quad V_D = 160 \text{ ml}$$

8. Ans. (B,C,D)

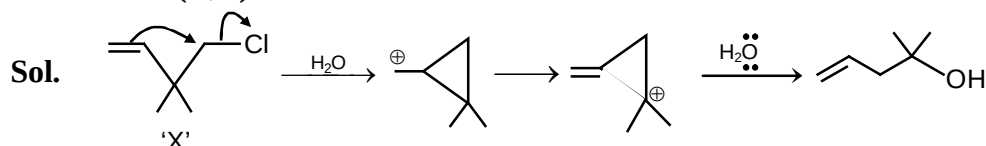
Sol. Aeration of pulp impregnates it thoroughly with bubbles. Xanthate anions are adsorbed on the surface of galena particles which makes them hydrophobic and they attach with air bubbles. Thus, the effective density of galena particles is drastically reduced. This causes galena to buoy up in spite of the fact that it is, in reality, much heavier than the siliceous gangue. Silica particles are wetted by the polar water molecules.

9. Ans. (A,B,D)



All other species on gaining electron release energy.

10. Ans. (A,B)



11. **Ans. (A,C,D)**

Sol. (A) This is because the base although not involved in RDS, does control how much carbanion is formed to take part in rate limiting step.

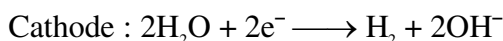
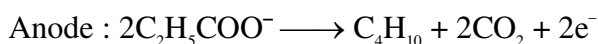
12. **Ans. (A,C,D)**

Sol. In case of (B) you would have made one half of a mole of alkane with one half of your alkyne left over.

SECTION-II

1. **Ans. (8.00)**

Sol. During Kolbe's electrolysis of potassium propanoate



$$\therefore \text{By } 4\text{F charge total moles of gases produced} = \frac{2}{(\text{C}_4\text{H}_{10})} + \frac{4}{(\text{CO}_2)} + \frac{2}{(\text{H}_2)}$$

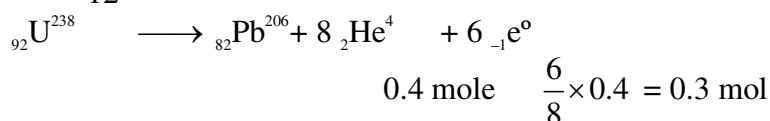
2. **Ans. (400.00)**

Sol. $(P_{\text{air}})_{\text{initial}} = 1 \text{ atm}$

$$(P_{\text{air}})_{\text{final}} = 1/3 \text{ atm} \quad (\because \text{volume is tripled})$$

$$P_{\text{He}} = 2/3 \text{ atm}$$

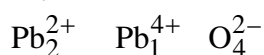
$$n_{\text{He}} = \frac{\frac{2}{3} \times 15}{\frac{1}{12} \times 300} = 0.4$$



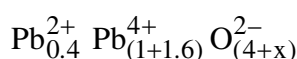
Energy required to produce 0.3 mole of electrons = $13.6 \times 0.3 \times N_A \times 1.6 \times 10^{-19} \times 10^{-3} \text{ kJ} \approx 400 \text{ kJ}$

3. **Ans. (43.00)**

Sol. Red lead: Pb_3O_4 ($2\text{PbO} \cdot \text{PbO}_2$)



↓ By aerobic oxidation



Applying charge balancing

$$0.8 + 4(1 + 1.6) = 2(4 + x)$$

$$11.2 + 8 + 2x$$

$$2x = 3.2$$

$$x = 1.6$$

So the formula is $\text{Pb}_3\text{O}_{5.6}$ or $\text{Pb}_{15}\text{O}_{28}$

4. **Ans. (3.00)**

5. **Ans. (7.00)**

(i, ii, iii, vi, viii, ix, x)

Sol. Fact Based

6. **Ans. (8.00)**

(a, b, c, d, e, f, i & j)

Sol. In carbene $:\text{CCl}_2$ singlet carbene is more stable due to $p\pi$ - $p\pi$ bonding overlapping between vacant p of carbon with filled orbital of Cl.

PART-3 : MATHEMATICS**SOLUTION****SECTION-I****1. Ans. (B)**

Sol. Required probability = $\frac{P(P_1 \text{ reaches final} \cap P_2 \text{ reaches semifinal but not in final})}{P(P_2 \text{ reaches semifinal but not in final})}$

$$= \frac{P(P_1 \text{ and } P_2 \text{ reaches semifinal} \cap P_1 \text{ wins semifinal but } P_2 \text{ loses semifinal})}{P(P_2 \text{ reaches semifinal and } P_2 \text{ loses semifinal})}$$

$$= \frac{\frac{{}^{14}C_2}{{}^{16}C_4} \times \left(\frac{1}{3}\right) \left(\frac{1}{2} + \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2}\right)}{\frac{{}^{15}C_3}{{}^{16}C_4} \times \frac{1}{2}} = \frac{2}{15}$$

2. Ans. (D)**Sol.** Since both the curves touches each other

$$\Rightarrow m_1 = m_2$$

$$4(x-a) = 2e^{2x}$$

$$4(x-a) = 2.2(x-a)^2$$

$$\Rightarrow (x-a)(x-a-1) = 0$$

$$\Rightarrow x = a \text{ or } x = a+1$$

If $x = a$ then (1st) curve $\Rightarrow y = 0$ & 2nd curve $\Rightarrow y = e^{2a}$ but $e^{2a} \neq 0 \forall a \in \mathbb{R}$

$$\therefore x \neq a$$

$$\Rightarrow x = a+1$$

$$\therefore 1^{\text{st}} \text{ curve} \Rightarrow y = 2 \quad \therefore (x, y) = (a+1, 2)$$

(x, y) lies on 2nd curve also, $\therefore 2 = e^{2a+2}$

$$\Rightarrow 2a+2 = \ln 2$$

$$a = \frac{1}{2}(\ln 2 - 2)$$

$$\Rightarrow -1 < a < 0$$

3. Ans. (B)**Sol.** Let $P(t, t^2)$ be the parametric co-ordinates of any point on the parabola $y = x^2$

$$\Rightarrow \left. \frac{dy}{dx} \right|_{(t, t^2)} = 2t$$

$$\text{Hence normal at P is } (y - t^2) = \frac{-1}{2t} (x - t)$$

If this line meets the curve again at $Q(t', t'^2)$, then

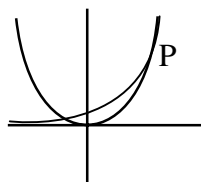
$$(t'^2 - t^2) = \left(-\frac{1}{2t}\right)(t' - t) \Rightarrow t' + t = -\frac{1}{2t} \Rightarrow t' = -t - \frac{1}{2t}$$

If ℓ is length of the normal chord then $z = (\ell)^2 = PQ^2$

$$\Rightarrow z = (t - t')^2 [1 + (t + t')^2] \Rightarrow z = \left(t + t + \frac{1}{2t}\right)^2 \left(1 + \left(-\frac{1}{2t}\right)^2\right) \Rightarrow z = \frac{(4t^2 + 1)^3}{16t^4}$$

Put $4t^2 = p$ where $p > 0$

$$z = \frac{(p+1)^3}{p^2} = (p+3) + \frac{3}{p} + \frac{1}{p^2} \Rightarrow \frac{dz}{dp} = \left(1 - \frac{3}{p^2}\right) - \frac{2}{p^3} = 0$$



$$\Rightarrow (p+1)^2 (p-2) = 0$$

$$\Rightarrow p = 2 \text{ as } p \neq -1 \text{ and } \frac{d^2z}{dp^2} = \frac{6}{p^3} + \frac{6}{p^4} > 0$$

Hence z is minimum when $p = 2 = 4t^2$

4. **Ans. (A)**

Sol. Number of numbers from 1 to N , coprime with $N =$

$$\left(N - \left(\frac{N}{2} + \frac{N}{3} + \dots + \frac{N}{11} \right) + \left(\frac{N}{2 \times 3} + \frac{N}{2 \times 5} + \dots + \frac{N}{7 \times 11} \right) - \sum \frac{N}{2.3.5} + \sum \frac{N}{2.3.5.7} - \sum \frac{N}{2.3.5.7.11} \right)$$

$$= N \left(1 - \frac{1}{2} \right) \left(1 - \frac{1}{3} \right) \dots \left(1 - \frac{1}{11} \right)$$

$$= 1 \times 2 \times 4 \times 6 \times 10 = 480$$

$$\Rightarrow P(\text{a coprime with } N) = \frac{480}{2 \times 3 \times 5 \times 7 \times 11} = \frac{16}{77}$$

5. **Ans. (A,B,C,D)**

Sol. $P_1S_2 + P_1S_1 = 2a$

$$|P_1S_2 - P_1S_1| = 2c$$

$\Rightarrow$ either $P_1S_2 = a + c$ and $P_1S_1 = a - c$ or $P_1S_2 = a - c$ and $P_1S_1 = a + c$

$$\Rightarrow P_1S_1 \times P_1S_2 = a^2 - c^2$$

Now $a^2 - b^2 = c^2 + d^2$ { C_1 and C_2 are confocal }

$$e_1e_2 = \sqrt{\frac{a^2 - b^2}{a^2}} \sqrt{\frac{c^2 + d^2}{c^2}} = \frac{a^2 - b^2}{ac} \quad \dots (i)$$

$$\text{Also } \frac{b^2}{a} = \frac{d^2}{c} = \frac{b^2 + d^2}{a + c} = \frac{a^2 - c^2}{a + c} = a - c$$

$$\Rightarrow b^2 = a^2 - ac \Rightarrow a^2 - b^2 = ac$$

$$\Rightarrow \frac{a^2 - b^2}{ac} = 1 \quad \dots (ii)$$

from (i) and (ii) $e_1e_2 = 1$

$$\text{because } a^2 - c^2 = b^2 + d^2 \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{c^2} = 1 \text{ and } \frac{x^2}{b^2} - \frac{y^2}{d^2} = 1$$

are also confocal

C_1 and C_2 intersect orthogonally (property)

6. **Ans. (A,C)**

Sol. $\arg(3 + 2x + |z|i) = \frac{\pi}{4}$

$$\frac{\sqrt{x^2 + y^2}}{2x + 3} = 1 \quad \left(x > -\frac{3}{2} \right)$$

$$\Rightarrow x^2 + y^2 = 4x^2 + 9 + 12x$$

$$\Rightarrow 3(x + 2)^2 - y^2 = 3$$

$$\Rightarrow \frac{(x + 2)^2}{1} - \frac{y^2}{3} = 1$$

$$\text{LR.} = \frac{2b^2}{a} = 6$$

7. **Ans. (A,C,D)**

any point on the given line will be

$$(2\lambda + 1, -3\lambda - 1, \lambda) \equiv (\alpha, \beta, \gamma)$$

$$\alpha + \beta + \gamma = 0$$

$$\text{Now } T_{r+1} = {}^9C_r (2x^2)^{9-r} (-x)^{-r}$$

$$= (-1)^r {}^9C_r 2^{9-r} x^{18-3r}$$

$$18 - 3r = 0 \Rightarrow r = 6$$

$$\therefore \text{coefficient of } x^0 = {}^9C_6 2^3 = \frac{9 \cdot 8 \cdot 7}{6} \cdot 8$$

8. **Ans. (A,B,C,D)**

Sol. Because $|z| \leq 1$ and $|iw| \leq 1$ and $|z - iw| = 2$, so z, iw are end points of diameter of $x^2 + y^2 = 1$.

Similarly $\bar{z}$ and $\overline{iw}$ are also end points of same diameter (because $|z - \overline{iw}| = 2$)

$\Rightarrow iw = \overline{iw} \Rightarrow iw$ lies on real axis

$\Rightarrow z$ lies on real axis

$\Rightarrow$ either $z = 1, iw = -1$ or $z = -1, iw = 1$

9. **Ans. (A,C,D)**

Sol. Obviously $f(1 + \alpha) = f(1 - \alpha) \forall \alpha \in (0, 1)$

Domain of f is $(0, 2)$

Hence (A) is true.

Maxima of $\ln(2x - x^2)$ as well as that of $\sin\left(\frac{\pi x}{2}\right)$ occur at $x = 1$.

Hence (C) is true

As $x \rightarrow 0^+$ or $x \rightarrow 2^-$, $f(x) \rightarrow -\infty$.

Hence (D) is true.

10. **Ans. (C,D)**

Sol.
$$I(k) = \int_0^\infty \frac{\ln x}{x^2 + kx + k^2} dx$$

put $x = \frac{k}{t}$ we get

$$I(k) = \frac{1}{k} \int_0^\infty \frac{\ln k - \ln t}{t^2 + t + 1} dt$$

$$\Rightarrow I(k) = \frac{\ln k}{k} \int_0^\infty \frac{dt}{t^2 + t + 1} - \frac{I(1)}{k}$$

$$\text{Now } I(1) = -I(1)$$

$$\Rightarrow I(1) = 0$$

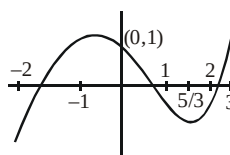
$$I(k) > \frac{\ln k}{k} \int_0^1 \frac{dt}{t^2 + t + 1} \quad (k > 1)$$

11. **Ans. (B,D)**

Sol. Clearly,
$$\begin{vmatrix} x & x & 5 \\ 1 & x & 0 \\ 1 & 1/5 & x \end{vmatrix} = 0 \Rightarrow x^3 - x^2 - 5x + 1 = 0$$

$$\text{Assume } f(x) = x^3 - x^2 - 5x + 1$$

$$\therefore f'(x) = 3x^2 - 2x - 5 = 0 \Rightarrow x = -1 \text{ \& } \frac{5}{3}$$



Clearly $[\alpha] + [\beta] + [\gamma] = -2 + 0 + 2 = 0$

Now $x^3 - x^2 - 5x + 1 = (x - \alpha)(x - \beta)(x - \gamma)$

Put $x = -1$, we get

$$(1 + \alpha)(1 + \beta)(1 + \gamma) = -(-1 - 1 + 5 + 1) = -4$$

Now, volume of parallelepiped

$$= |x^3 - x^2 - 5x + 1| \quad x \in [-2, 3]$$

Maximum value will occur at $x = \frac{5}{3}$ and is equal to

$$\left| \frac{125}{27} - \frac{25}{9} - \frac{25}{3} + 1 \right| = \left| \frac{125 - 75 - 225 + 27}{27} \right| = \frac{148}{27}$$

12. **Ans. (A,C,D)**

Sol. $f(x + y) = 2^x f(y) + 4^y f(x)$

replace x into y and y into x , we get

$$f(x + y) = 2^y f(x) + 4^x f(y) \Rightarrow 2^x f(y) + 4^y f(x) = 2^y f(x) + 4^x f(y)$$

$$\Rightarrow \frac{f(x)}{4^x - 2^x} = \frac{f(y)}{4^y - 2^y} = k \Rightarrow f(x) = k(4^x - 2^x)$$

since, $f'(0) = \ln 2$, we get $k = 1$

Hence, $f(x) = 4^x - 2^x$

SECTION-II

1. **Ans. (87.00)**

Sol. If child buys pencil on the first day then the remaning $n-1$ rupees can be spend in S_{n-1} ways and if he buys pen on the first day then the remaning $n-2$ rupees can be spend in S_{n-2} ways similarly S_{n-2} ways for book on the first day.

$\therefore$ the recurrence relation is

$$S_n = S_{n-1} + S_{n-2} + S_{n-2}$$

$$\therefore S_n = S_{n-1} + 2 S_{n-2}$$

Clearly $S_1 = 1, S_2 = 3\{(1,1), 2, 2\}$

$$\text{Now, } S_3 = S_2 + 2S_1 = 5$$

$$S_4 = 5 + 2 \cdot 3 = 11$$

$$S_5 = 11 + 2 \times 5 = 21$$

$$S_6 = 21 + 22 = 43$$

$$S_7 = 43 \times 2 \times 21$$

$$= 43 + 42$$

$$= 85$$

$$\therefore \left(\frac{S_n - S_{n-1}}{S_{n-2}} + S_7 \right) = 85 + 2 = 87$$

2. **Ans. (144.00)**

Sol. Common root of $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ is 1

$$\Rightarrow 1 + a + b = 0 \quad \dots\dots(i)$$

Now roots of $x^2 + ax + b = 0$ are 1, b and roots of $x^2 + bx + a = 0$ are 1, a

hence roots of $x^3 + px^2 + qx + 72$ are 1, a , b

$$\Rightarrow ab = -72 \quad \dots\dots(ii)$$

from (i) and (ii) $\Rightarrow (a, b) = (8, -9)$ or $(-9, 8) \Rightarrow f(x) = (x-1)(x-8)(x+9)$

$$\Rightarrow f(-1) = (-2)(-9)(8) = 144$$

3. Ans. (150.00)

Sol. $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$
 $= 2(|z_1|^2 + |z_2|^2 + |z_3|^2) - (z_1\bar{z}_2 + z_2\bar{z}_1 + z_1\bar{z}_3 + z_3\bar{z}_1 + z_3\bar{z}_2 + z_2\bar{z}_3)$
 $= 100 - (z_1\bar{z}_2 + z_2\bar{z}_1 + z_1\bar{z}_3 + z_3\bar{z}_1 + z_3\bar{z}_2 + z_2\bar{z}_3)$
 Now, $|z_1 + z_2 + z_3|^2 \geq 0 \Rightarrow 50 + (z_1\bar{z}_2 + z_2\bar{z}_1 + z_1\bar{z}_3 + z_3\bar{z}_1 + z_3\bar{z}_2 + z_2\bar{z}_3) \geq 0$
 $\Rightarrow -(z_1\bar{z}_2 + z_2\bar{z}_1 + z_1\bar{z}_3 + z_3\bar{z}_1 + z_3\bar{z}_2 + z_2\bar{z}_3) \leq 50 \Rightarrow \text{max value} = 100 + 50 = 150$

4. Ans. (5.00)

Sol. Given $f(x+5) \geq f(x) + 5 \dots (1)$

$$f(x+1) \leq f(x) + 1 \dots (2)$$

from (1) and (2), we can write

$$\begin{aligned} f(x) + 5 &\leq f(x+5) \leq f(x+4) + 1 \\ &\leq f(x+3) + 2 \\ &\leq f(x+2) + 3 \\ &\leq f(x+1) + 4 \\ &\leq f(x) + 5 \end{aligned}$$

$\therefore$ equality exist everywhere

$$\therefore f(x+1) = f(x) + 1 \quad \because f(1) = 1$$

$$\Rightarrow f(2) = 2, f(3) = 3, \dots, f(2013) = 2013.$$

$$\text{Now } g(x) = f(x) + 5 - x \Rightarrow g(2013) = 5$$

5. Ans. (4.00)

Sol. Equation of tangent at (x_i, y_i) on the curve $y = f(x)$ is : $Y - y_i = m(X - x_i)$

$$\text{Put } Y = 0, X = x_i - \frac{y_i}{m} \Rightarrow x_{i+1} = x_i - \frac{y_i}{m}$$

$$\therefore \frac{x_{i+1}}{x_i} - 1 = -\frac{y_i}{mx_i}$$

If $x_1, x_2, x_3, \dots, x_n$ form G.P. then $r - 1 = -\frac{y_i}{mx_i}$ (where r is common ratio of G.P.)

$$\therefore m = \frac{dy}{dx} = -\frac{y}{(r-1)x} \Rightarrow \frac{dy}{y} + \frac{dx}{(r-1)x} = 0$$

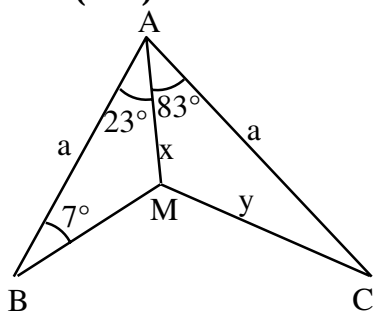
$$\Rightarrow \ell n y + \left(\frac{1}{r-1}\right) \ell n x = k$$

$$\Rightarrow y(x)^{\frac{1}{r-1}} = c$$

$$\Rightarrow r = 2, xy = c, e = \sqrt{2}$$

6. Ans. (0.50)

Sol.



In $\triangle ABM$

$$\frac{x}{\sin 7^\circ} = \frac{a}{\sin 150^\circ} \Rightarrow x = 2a \sin 7^\circ \quad \dots(i)$$

In $\triangle AMC$,

$$\cos 83^\circ = \frac{x^2 + a^2 - y^2}{2ax} \Rightarrow y^2 = -2a(2a \sin 7^\circ)(\cos 83^\circ) + (2a \sin 7^\circ)^2 + a^2$$

$$\Rightarrow y^2 = a^2 \Rightarrow \angle AMC = \angle MAC = 83^\circ$$

$$\Rightarrow \angle ACM = 14^\circ$$

Now, $\angle ABC = \angle ACB = 37^\circ$

$$\Rightarrow \angle MBC = 30^\circ \Rightarrow \sin (\angle MBC) = \frac{1}{2} = 0.50$$

 ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)	CLASSROOM CONTACT PROGRAMME (Academic Session : 2020 - 2021)	JEE(Advanced) FULL SYLLABUS 21-08-2021
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JEE(Main+ Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH
PAPER-2
PART-1 : PHYSICS
SOLUTION
SECTION-I
1. Ans. (A, D)
Sol. Potential difference between two points at distance $2R$ and R is

$$\Delta V = \frac{\lambda}{2\pi\epsilon_0} \ln 2$$

So

$$\frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 = \frac{\lambda Q}{2\pi\epsilon_0} \ln 2$$

$$\therefore v = \sqrt{v_0^2 + \frac{\lambda Q}{\pi\epsilon_0 m} \ln 2}$$

Also,

 If v_t is the component $\perp$ to radius then according to angular momentum conservation about line charge,

$$mv_t(2R) = mv_0R$$

$$\therefore v_t = \frac{v_0}{2}$$

&

$$v_t^2 + v_r^2 = v^2$$

$$\therefore v_r^2 = v^2 - v_t^2$$

$$= v_0^2 + \frac{\lambda Q}{\pi\epsilon_0 m} \ln 2 - \frac{v_0^2}{4}$$

$$\therefore v_r = \sqrt{\frac{3v_0^2}{4} + \frac{\lambda Q}{\pi\epsilon_0 m} \ln 2}$$

2. Ans. (B, D)
Sol. $a_A = g$, upward

$$a_B = 0$$

 time to collide, $t = \frac{H}{u \sin \theta}$; for motion of B

$$H = \frac{1}{2}a_A t^2$$

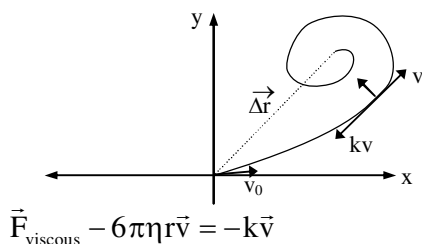
$$H = \frac{1}{2} \times g \times \left(\frac{H}{u \sin \theta} \right)^2$$

$$\Rightarrow u = 10 \text{ m/s}$$

$$V = \frac{H}{\tan \theta \cdot t} = u \cos \theta = 5\sqrt{2} \text{ m/s}$$

3. Ans. (B, C)

Sol.



$$\vec{F}_{\text{viscous}} - 6\pi\eta r\vec{v} = -k\vec{v}$$

and $\vec{F} = \vec{A} \times \vec{v}$ is always perpendicular to $\vec{F}$

$$\therefore \frac{m d\vec{v}}{dt} = \vec{A} \times \vec{v} - k\vec{v}$$

$$\therefore \int m d\vec{v} = \int \vec{A} \times d\vec{r} - \int k d\vec{r}$$

$$\therefore m(-\vec{v}_i) = \vec{A} \times \vec{\Delta r} - k\vec{\Delta r}$$

$$\therefore \Delta r = \frac{mv_0}{\sqrt{A^2 + k^2}} = \frac{mv_0}{\sqrt{A^2 + (6\pi\eta r)^2}}$$

$$\& \frac{dv}{dt} = -\frac{k}{m}v$$

$$\therefore \int_{v_0}^{\frac{v_0}{2}} \frac{dv}{v} = -\frac{k}{m} \int_0^t dt$$

$$\therefore \ln \frac{1}{2} = -\frac{k}{m}t$$

$$\therefore t = \frac{m}{k} \ln 2$$

$$= \frac{m \ln 2}{6\pi\eta r}$$

4. Ans. (A, D)

Sol. Since $\rho \propto \frac{1}{r} \Rightarrow \rho = \frac{a}{r}$

$$\text{And } M = \int \left(\frac{a}{r} \right) 4\pi r^2 dr = 2\pi a R^2$$

$$\therefore a = \frac{M}{2\pi R^2}$$

$$\text{Now } V_c = -Ga \int_0^R 4\pi dr = -Ga4\pi R = -\frac{GM}{2\pi R^2} \times 4\pi R = -\frac{2GM}{R}$$

$$\& \quad E(r) = \frac{a \int_0^r 4\pi r'^2 dr'}{r^2} = 2\pi a G$$

Now, self energy is equal to gravitational field energy.

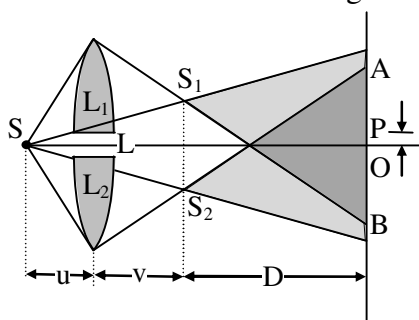
$$U_{0 \rightarrow R} = -\frac{1}{2} \times \left(\frac{1}{4\pi G} \right) (2\pi a G)^2 \frac{4}{3} \pi R^3 = -\frac{GM^2}{6R}$$

$$\& \quad U_{R \rightarrow \infty} = -\frac{GM^2}{2R}$$

$$\therefore \quad U = -\frac{2}{3} \frac{GM^2}{R}$$

5. **Ans. (A, C)**

Sol. (i) As shown in figure each part of the lens will form image of S which will act as coherent sources.



From lens equation, we can write

$$\frac{1}{v} - \frac{1}{-15} = \frac{1}{10}$$

$$\text{or } v = 30 \text{ cm}$$

$$m = \frac{-v}{u} = -2$$

$$\text{Also, } d = 3 \times 0.5 \text{ mm} = 1.5 \text{ mm}$$

$$D = 1.30 - 0.30 = 1 \text{ m}$$

Now from the theory of interference the distance y of a point P on the screen is given by

$$y = \frac{D}{d} (\Delta x)$$

And as point is third maximum

$$\Delta x = 3\lambda$$

$$\text{So, } y = \frac{D}{d} (3\lambda)$$

$$\text{or } y = \frac{5 \times 10^{-7}}{0.5 \times 10^{-3}} \times 10^{-3} \text{ m} = 1 \text{ mm}$$

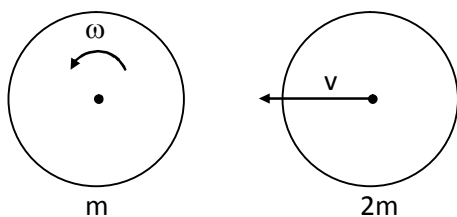
6. **Ans. (A, C)**

$$\text{Sol. } m \frac{dv}{dt} = -B_0 a I$$

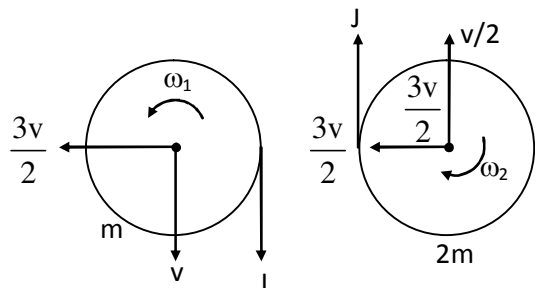
$$L \frac{dI}{dt} = -B_0 a v \Rightarrow \frac{d^2 v}{dt^2} + \omega^2 v = 0 \Rightarrow v = v_0 \cos \omega t \text{ and } s = \frac{v_0}{\omega} \sin \omega t$$

7. Ans. (A, D)

Sol. Before collision



After collision



$$\frac{v}{2} + \omega_2 r = \omega_1 r - v \quad \dots(1)$$

$$J = mv \quad \dots(2)$$

$$Jr = \left(\frac{2mr^2}{2} \right) \omega_2 \quad \dots(3)$$

$$\frac{mr^2}{2} \omega - Jr = \frac{mr^2}{2} \omega_1 \quad \dots(4)$$

$$\text{So, } \omega = 2\omega_2 + \omega_1$$

$$\frac{3v}{2r} = \omega_1 - \omega_2$$

$$\text{i.e. } \frac{3v}{2R} \left(\frac{J}{m} \right) = \omega_1 - \omega_2$$

$$\frac{3}{2r} \left(\frac{mr\omega^2}{m} \right) = \omega_1 - \omega_2$$

$$\therefore \frac{5}{2} \omega_2 = \omega_1$$

$$\text{Solving } \omega_2 = \frac{2\omega}{9} \text{ and } \omega_1 = \frac{5\omega}{9} \text{ and } v = \frac{2\omega r}{9}$$

8. Ans. (B, D)

Sol. $K_{\max} = \frac{hC}{\lambda_{\min}} - \phi$

$$\text{For A} \Rightarrow K_{\max} = 3.1 - 1.55 = 1.55 \text{ eV}$$

$$\text{For B} \Rightarrow K_{\max} = 3.1 - 2.48 = 0.62 \text{ eV}$$

In experiment with metal A complete spectrum is able to eject photo electrons but for metal B spectrum from 400nm to 500nm, will be able to eject photo electrons. Number of photons in spectrum 400nm to 500nm will be less than 1/3 of total photons in white light.

9. Ans. (B)

Sol Since piston (II) is in equilibrium,

$$P_B = P_C$$

$$\& \text{ if } v' = \frac{41V_0}{25} \Rightarrow V_B = 2V_0 - \frac{41V_0}{25} = \frac{9V_0}{25}$$

process is adiabatic in section (B),

$$P_0 V_0^{\gamma} = (P_B) \left(\frac{9V_0}{25} \right)^{\gamma}$$

$$\therefore P_B = \left(\frac{125}{27} \right)^{P_0} = 4.6 P_0$$

So closest value is $5 P_0$

$$\& T_C = \frac{P_C V_C}{R} = \frac{\left(\frac{125}{27} \right) \left(\frac{41V_0}{25} \right)}{\left(\frac{25}{3} \right)} = \frac{41}{45} P_0 V_0 = T_A$$

$$\text{So, } P_A = \frac{2RT_A}{V_A} = 2 \times \frac{25}{3} \times \frac{41}{45} P_0 V_0 = 7.6 P_0$$

So closest value is $8 P_0$

Since Net work done by gases is zero,

$$\Delta Q_{\text{system}} = \Delta U_A + \Delta U_B + \Delta U_C = \frac{368}{9} P_0 V_0$$

10. Ans. (C)

Sol.

$\frac{14V_0}{9}$	V_0	$\frac{4V_0}{9}$
-------------------	-------	------------------

For (C) part,

$$P_0 V_0^{1.5} = P_f \left(\frac{4V_0}{9} \right)^{3/2}$$

$$\therefore P_f = \frac{27P_0}{8} = P_{3f} = P_{1f}$$

For (A) part,

$$\frac{P_0 V_0}{T_0} = \frac{\left(\frac{27P_0}{8} \right) \left(\frac{14V_0}{9} \right)}{T_{1f}}$$

$$\therefore T_{1f} = \frac{21T_0}{4}$$

For (B) part,

$$\frac{P_0 V_0}{T_0} = \left(\frac{P_{2f} V_0}{21T_0 / 4} \right)$$

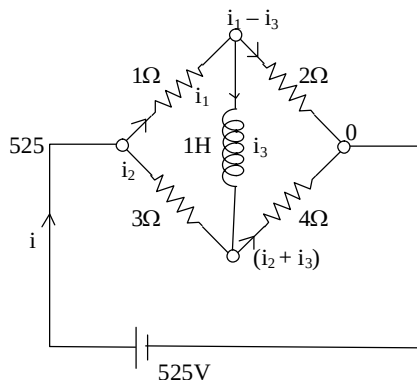
$$\therefore P_{2f} = \frac{21}{4} P_0$$

$$\therefore \Delta Q = \Delta U_A + \Delta U_B + \Delta U_C$$

$$= \frac{f}{2} \left(\frac{21}{4} P_0 V_0 + \frac{21}{4} P_0 V_0 + \frac{3}{2} P_0 V_0 - 3 P_0 V_0 \right) = 18 P_0 V_0$$

11. Ans. (C)

Sol.



$$i_1 + \frac{di_3}{dt} = 3i_2 \quad \dots\dots\dots(1)$$

$$525 = i_1 + 2(i_1 - i_3) \quad \dots\dots\dots(2)$$

$$525 = 3i_2 + 4(i_2 + i_3) \quad \dots\dots\dots(3)$$

$$\frac{2i_3}{3} + 175 + \frac{di_3}{dt} = 225 - \frac{12i_3}{7}$$

$$\therefore 21 \frac{di_3}{dt} = 1050 - 50i_3$$

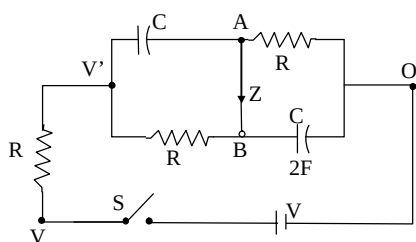
$$-\left(\frac{di_3}{i_3 - 21} \right) = 50 \frac{dt}{21}$$

$$\therefore \ell n(i_3 - 21) \Big|_0^{i_3} = -\frac{50t}{21}$$

$$\therefore i_3 = 21 \left(1 - e^{-\frac{50t}{21}} \right)$$

12. Ans. (A)

Sol.



$$\frac{V - V'}{R} = \frac{C}{2} \frac{d}{dt}(V') + \frac{V'}{2R}$$

$$\frac{2(V - V')}{2R} - \frac{V'}{2R} = \frac{C}{2} \frac{dV'}{dt}$$

$$\frac{2V - 3V'}{RC} = \frac{dV'}{dt}$$

$$\int_0^{V'} \frac{dV'}{3V' - 2V} = -\int_0^t \frac{dt}{RC}$$

$$\begin{aligned} \ln(3V' - 2V) \Big|_0^{V'} &= -\frac{3t}{RC} \\ \therefore \ln\left(\frac{2V - 3V'}{2V}\right) &= -\frac{3t}{RC} \\ \therefore V' &= \frac{2V}{3} \left(1 - e^{-\frac{3t}{RC}}\right) \& \frac{dV'}{dt} = \frac{2V}{RC} e^{-\frac{3t}{RC}} \\ \text{So, } \frac{C}{2} \frac{dV'}{dt} &= Z + \frac{V'}{2R} \\ \therefore \frac{V}{R} e^{-\frac{3t}{RC}} &= Z + \frac{V}{3R} \left(1 - e^{-\frac{3t}{RC}}\right) \\ \therefore Z &= \frac{4V}{3R} \left(e^{-\frac{3t}{RC}} - \frac{1}{4}\right) \end{aligned}$$

SECTION-II

1. **Ans. 8.00**

Sol. Let $\frac{5m}{4} = M$

$$E = 2\left(\frac{1}{2}mv_0^2\right) + \frac{1}{2}I\omega^2 + \frac{1}{2}mv_A^2 + mgr_0(1 - \cos\theta)$$

$$\& \text{ Since } v_0 = \omega r \& I = \frac{1}{2}Mr^2$$

$$\text{Also, } v_A^2 = (\omega r)^2 + (\omega r_0)^2 - 2\omega^2 r r_0 \cos\theta$$

$$v_A^2 = \omega^2(r - r_0)^2 \quad [\because \theta \text{ is small}]$$

Thus,

$$E = \frac{3}{2}M\omega^2 r^2 + \frac{1}{2}m(r - r_0)^2 \omega^2 + mgr_0(1 - \cos\theta) = \text{constant}$$

$$\therefore \frac{dE}{dt} = 3Mr^2\omega\alpha + m(r - r_0)^2\omega\alpha + mgr_0(\theta)\omega = 0$$

$$\therefore \alpha = \left(\frac{-mgr_0}{3Mr^2 + m(r - r_0)^2}\right)\theta$$

$$= -\left(\frac{g}{8r}\right)\theta$$

2. **Ans. 3.00**

Sol. $E = \frac{(d/2)^2}{2(L/2)} \left(\frac{dB}{dt}\right)$

Now $\int \tau dt = I\omega$

$$\therefore \int 2 \times qE \left(\frac{L}{2}\right) dt = \left(\frac{mL^2}{4} \times 2\right) \omega$$

$$\therefore qL \left(\frac{d^2}{4L} B_0\right) = \frac{mL^2}{2} \omega$$

$$\therefore \omega = \frac{qB_0 d^2}{2mL^2}$$

$$\therefore x = 1 ; y = 2 \Rightarrow x + y = 3.$$

3. Ans. 40.00

Sol. The deviation suffered by the ray, incident at angle i is

$$\phi = (i - r) + (180^\circ - 2r) + (i - r)$$

$$= 180^\circ - 4r + 2i$$

$$\therefore \delta = 180^\circ - \phi = 4r - 2i$$

$$\delta \text{ is maximum when } \frac{d\delta}{di} = 0$$

$$\Rightarrow 4 \frac{dr}{di} - 2 = 0$$

$$\Rightarrow \frac{dr}{di} = \frac{1}{2}$$

....(1)

Snell's law

$$1. \sin i = \mu \sin r$$

$$\sin i = \frac{4}{3} \sin r$$

$$\sin r = \frac{3}{4} \sin i$$

$$\Rightarrow \cos r \frac{dr}{di} = \frac{3}{4} \cos i$$

$$\therefore \text{ from (1) } \frac{3 \cos i}{4 \cos r} = \frac{1}{2}$$

$$3 \cos i = 2 \cos r$$

$$3 \cos i = 2 \sqrt{1 - \sin^2 r}$$

$$3 \cos i = 2 \sqrt{1 - \frac{9}{16} \sin^2 i} \quad [\text{using (2)}]$$

$$(6 \cos i)^2 = 16 - 9 \sin^2 i$$

$$36(1 - \sin^2 i) = 16 - 9 \sin^2 i$$

$$27 \sin^2 i = 20$$

$$\Rightarrow \sin i = \sqrt{\frac{20}{27}}; i = 60^\circ$$

$$\text{From (2) } \sin r = \frac{3}{4} \sqrt{\frac{20}{27}}$$

$$= \sqrt{\frac{5}{12}} \Rightarrow r = 40^\circ$$

$$\therefore \delta_{\max} = 4r - 2i = 4 \times 40 - 2 \times 60 = 40^\circ$$

4. Ans. 12.00

Sol. Time required for evaporation to reduce the volume to half is $t_{1/2} = 24$ hr.

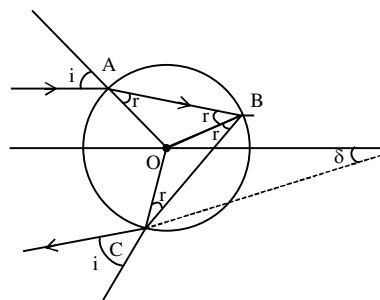
$$\therefore 1 = \frac{\ell \ln 2}{t_{1/2}} = \frac{\ell \ln 2}{24}$$

Volume added per hour to the pot = nv_0

$$\therefore \frac{dv}{dt} = nv_0 - \lambda v$$

$$\int_{v=0}^v \frac{dv}{nv_0 - \lambda v} = \int_0^t dt \Rightarrow [\ell \ln(nv_0 - \lambda v)]_{v=0}^v = -\lambda t$$

$$\ell \ln \left(\frac{nv_0 - \lambda v}{nv_0} \right) = -\lambda t \Rightarrow 1 - \frac{\lambda v}{nv_0} = e^{-\lambda t}$$



$$v = \frac{nv_0}{\lambda} (1 - e^{-\lambda t})$$

$$t = 24 \text{ hr} ; \lambda t = \ell n 2$$

$$\begin{aligned} v &= \frac{24nv_0}{\ell n 2} [1 - e^{-\lambda n 2}] \\ &= 24 \frac{nv_0}{\ell n 2} \left[1 - \frac{1}{2} \right] = \frac{12}{\ell n 2} nv_0 \end{aligned}$$

5. **Ans. 3.12**

Sol. $(180)^2 + (255 - 15t)^2 = (300 t)^2$

$$\therefore t = 1 \text{ s}$$

$$\text{Also } f' = \left(\frac{300 + 15 \cos 37}{300 - 10 \cos 353} \right) f_0 = \frac{312}{294} \times 2.94 = 3.12 \text{ Hz.}$$

6. **Ans. 28.28**

Sol. $-h = u \sin \theta t - \frac{1}{2} g t^2$

$$\therefore \frac{g}{2} t^2 - h = u \sin \theta t \quad \dots\dots\dots (1)$$

$$R = u \cos \theta t \quad \dots\dots\dots (2)$$

$$\therefore \frac{g^2 t^4}{2} + h^2 - g h t^2 + R^2 = u^2 t^2 \quad \dots\dots\dots (3)$$

$$\therefore g^2 t^3 - 2 g h t + 2 R \frac{dR}{dt} = 2 u^2 t \left[\frac{dR}{dt} = 0 \right]$$

$$\therefore g^2 t^2 - 2 g h = 2 u^2$$

$$\therefore g^2 t^2 = 6 g h$$

$$\therefore t^2 = \frac{6 h}{g}$$

Substituting t^2 in (3),

$$\therefore \frac{g^2}{4} \left(\frac{36 h^2}{g^2} \right) + h^2 - g h \left(\frac{6 h}{g} \right) + R^2 = (2 g h) \left(\frac{6 h}{g} \right)$$

$$\therefore R^2 = 8 h^2$$

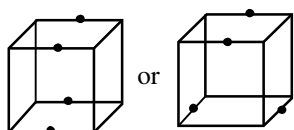
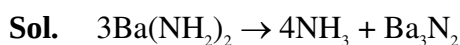
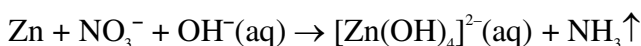
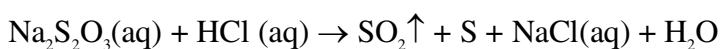
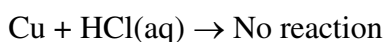
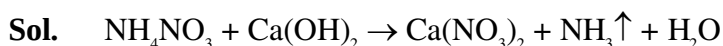
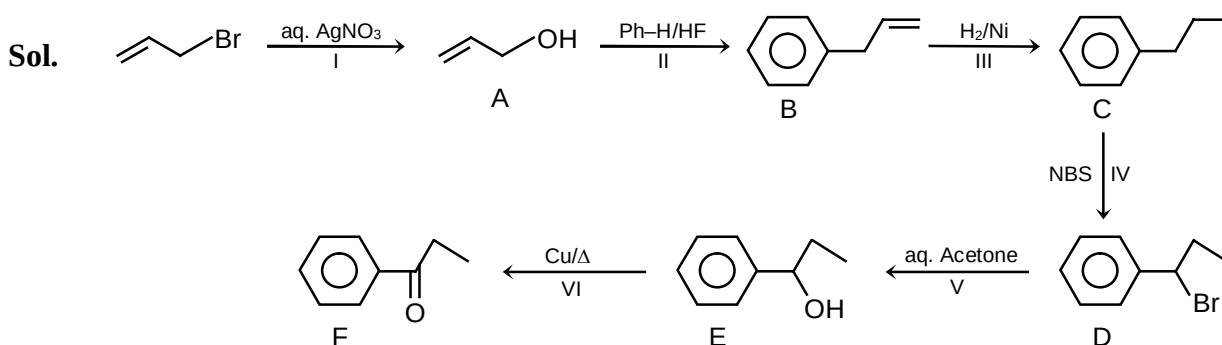
$$\therefore R = 2 \sqrt{2} h$$

$$\therefore \frac{R}{h} = 2 \sqrt{2}$$

$$\text{Hence } \frac{10 R}{h} = 28.28$$

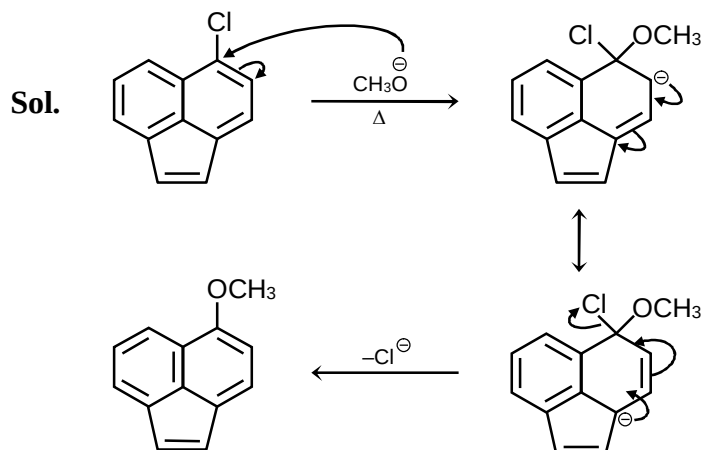
PART-2 : CHEMISTRY**SOLUTION****SECTION-I**1. **Ans.(A,B,D)****Sol.** $\Rightarrow$ In any compound of type AB, lattice of both cation & anion is same $\Rightarrow$ The shaded plane shows layer of ccp lattice, which does not contain any octahedral or tetrahedral void. $\Rightarrow$ In one unit cell $A = 4$ $B = 8$ (in all tetrahedral voids)

$$C = 4 \times \frac{1}{4} = 1$$

 $(\therefore \text{shared octahedral voids are present as edge centres so C can be placed as.})$ 2. **Ans.(B,D)****Sol.** Whenever 3 Bidentate legends are present in an octahedral complex, there is no plane of symmetry and hence the compound is optically active. $[\text{Ru}(\text{bipy})_3]^{4+}$, $[\text{Cr}(\text{ox})_3]^{3-}$ are optically active.3. **Ans. (B,C)**4. **Ans. (A,C,D)**5. **Ans.(B,C)**

6. Ans.(A,B,C)

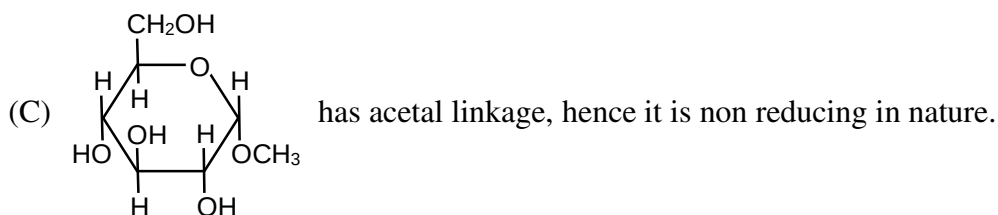
7. Ans. (B,C,D)



8. Ans. (C,D)

Sol. (A) Monosaccharides generally show mutarotation.

(B) D-fructose $\xrightarrow{\text{H}_2/\text{Ni}}$ sorbitol + mannitol



(D) Fructose and glucose are distinguished by Seliwanoff's method.

9. Ans. (B)

10. Ans.(D)

Sol. T_b = boiling point $T_m \Rightarrow$ Melting point
 $\Rightarrow$ Condensation at normal melting point & 1 atm is spontaneous



(1atm, T_m) (1atm, T_m)

$$\Delta S_{\text{system}} = \frac{\Delta H}{T_b} \Rightarrow -ve$$

$$\Delta S_{\text{surr}} = -\frac{\Delta H}{T_m} \Rightarrow +ve$$

$$\therefore T_m < T_b$$

$$|\Delta S_{\text{surr}}| > |\Delta S_{\text{system}}|$$

$$\Delta G < 0$$

$\Rightarrow$ Fusion at normal boiling point & 1 atm is spontaneous



(1 atm, T_b) (1atm, T_b)

$$\Delta S_{\text{system}} = \frac{\Delta H}{T_m} \Rightarrow +ve$$

$$\Delta S_{\text{surr}} = -\frac{\Delta H}{T_b} \Rightarrow -ve$$

$$|\Delta S_{\text{system}}| > |\Delta S_{\text{surr}}|$$

$$\Delta G < 0$$

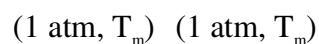
$\Rightarrow$ Vaporisation at triple point is reversible

At triple point, all 3 states solid, liquid gas are in eq^m with each other.

$$\text{So } \Delta S_{\text{system}} = -\Delta S_{\text{surr}} \quad \Delta H_{\text{vap}} \Rightarrow +ve$$

$$\Delta G = 0 \quad \Delta S_{\text{system}} > 0$$

$\Rightarrow$ Freezing at normal melting point & 1 atm is reversible



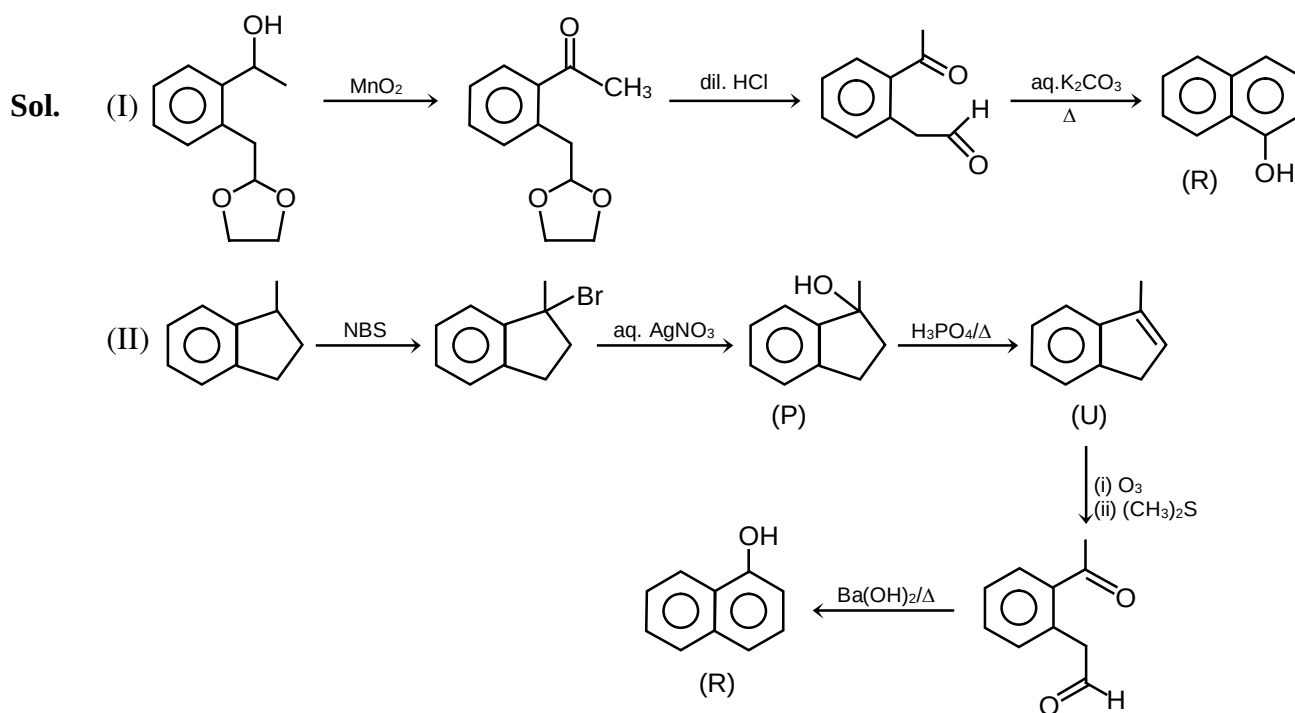
$$\Delta G = 0$$

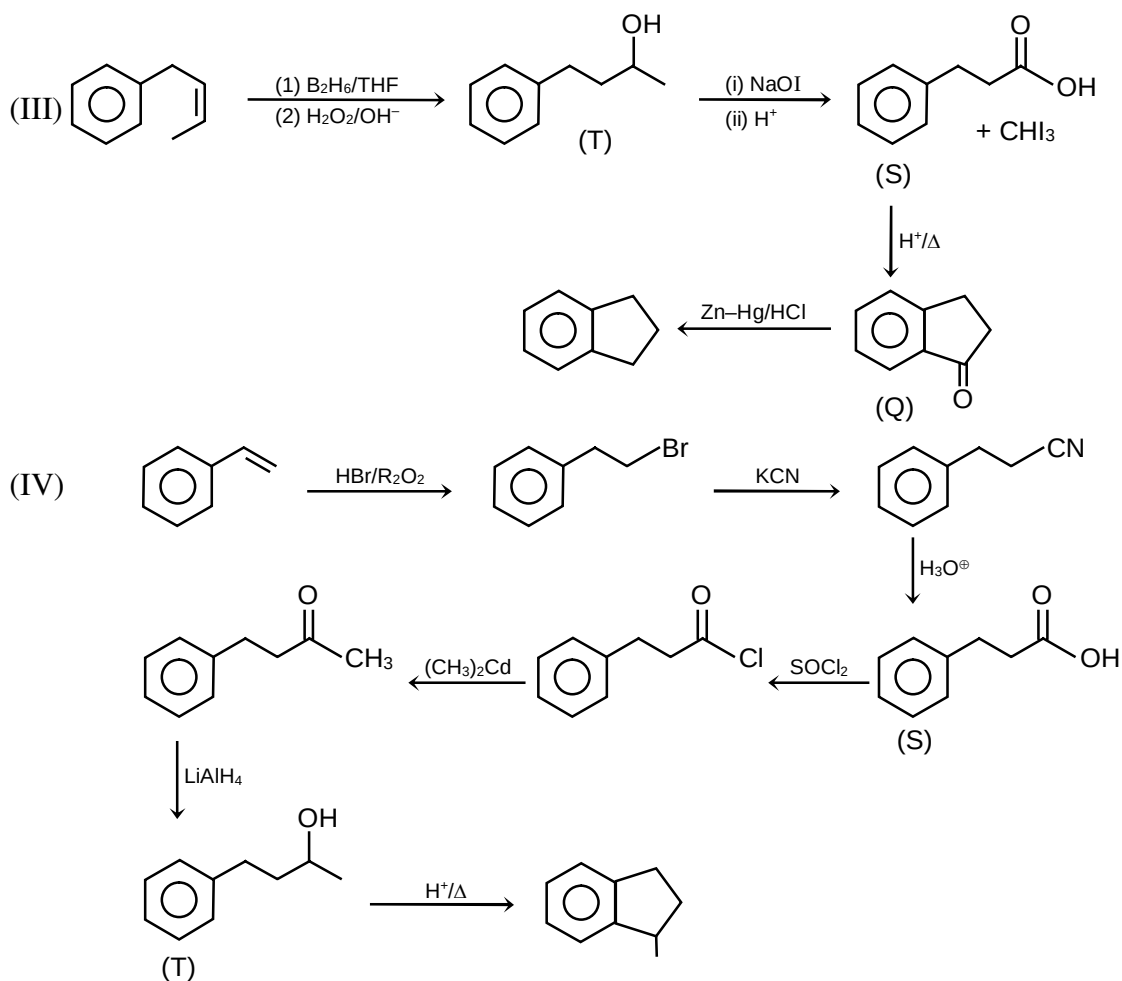
$$\Delta S_{\text{system}} < 0$$

$$|\Delta S_{\text{system}}| = |\Delta S_{\text{surr}}|$$

11. Ans. (C)

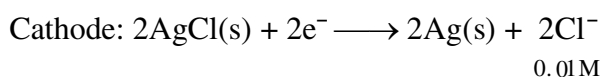
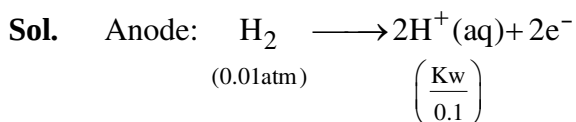
12. Ans. (B)





SECTION-II

1. Ans. (100.00)



$$E_{\text{Cell}}^0 = E_{\text{cathode}}^0 - E_{\text{anode}}^0 = 0.3$$

$$E_{\text{cell}} = E_{\text{Cell}}^0 - \frac{0.06}{2} \log \left(\frac{[\text{Cl}^-]^2 [\text{H}^+]^2}{P_{\text{H}_2}} \right)$$

$$1.02 = 0.3 - \frac{0.06}{2} \log \left[\frac{(0.01)^2 \left(\frac{K_w}{0.1} \right)^2}{(0.01)} \right]$$

$$\log K_w = -\frac{0.72}{0.06} = -12$$

$$K_w = 10^{-12}$$

2. **Ans. (8.00)**

Sol. Except S and PCl_5 , all other reagents can oxidise SO_2 .

Boiling aqueous solution of SO_2 with Sulphur forms thiosulphate, in which SO_2 is reduced.

$\text{PCl}_5 + \text{SO}_2 \longrightarrow \text{POCl}_3 + \text{SOCl}_2$. In this reaction SO_2 is not oxidised.

Aqueous Fe^{3+} oxidises SO_2 to sulphate. Cl_2 oxidises it to SO_2Cl_2 .

3. **Ans. (2.00)**

Sol. Process OA is isobaric

$$\text{so } q_{\text{OA}} = \Delta H = nC_{p,m} \Delta T$$

Process OB is $P = KV^{-n}$

$$PV^n = K$$

$$q_{\text{OB}} = nC_m \Delta T$$

$$\text{Where } C_m = C_{v,m} - \frac{R}{(n-1)}$$

$$\frac{q_{\text{OA}}}{q_{\text{OB}}} = \frac{nC_{p,m} \Delta T}{nC_m \Delta T} = \frac{C_{p,m}}{C_m} = 2$$

$$C_{p,m} = 2C_m$$

$$4R = 2 \left(3R - \frac{R}{n-1} \right)$$

$$\frac{1}{n-1} = 1$$

$$n = 2$$

4. **Ans. (375.00)**

Sol.

	$2R(g)$	$\rightarrow$	$S(g) + 3Q(g) + U(l)$
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Initial pressure	800mm	0	0	0
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At $t = 1$ hour	$800 - 2P$	P	$3P$	40mm
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$$800 + 2P + 40 = 800 + 640$$

$$2P + 40 = 640$$

$$= 200 \text{ mm}$$

Method –I for second order reaction $\frac{1}{(P_R)_t} - \frac{1}{(P_R)_0} = kt$

$$\frac{1}{200} - \frac{1}{800} = k \times 1, \quad k = \frac{3}{800} \text{ mm}^{-1} \text{ hour}^{-1}$$

$$\frac{1}{(P_R)_t} - \frac{1}{800} = k \times 5, \quad (P_R)_t \Rightarrow \text{Partial pressure of R left at time } t$$

$$(P_R)_t = 50 \text{ mm}$$

$\therefore$ Partial pressure of S formed at $t = 5$ hour = 375mm

Method –II For a second order reaction

$$800 \xrightarrow{(t_{1/2})_I} 400 \xrightarrow{(t_{1/2})_{II}} 200 \left((t_{1/2})_{II} = 2(t_{1/2})_I \right)$$

$$(t_{1/2})_{II} + (t_{1/2})_I = 60 \text{ min}$$

$$2(t_{1/2})_I + (t_{1/2})_I = 60 \text{ min}$$

$$(t_{1/2})_I = 20 \text{ min}$$

At $t = 5$ hour (300 min)

$800 \xrightarrow{20 \text{ min}} 400 \xrightarrow{40 \text{ min}} 200 \xrightarrow{80 \text{ min}} 100 \xrightarrow{160 \text{ min}} 50$

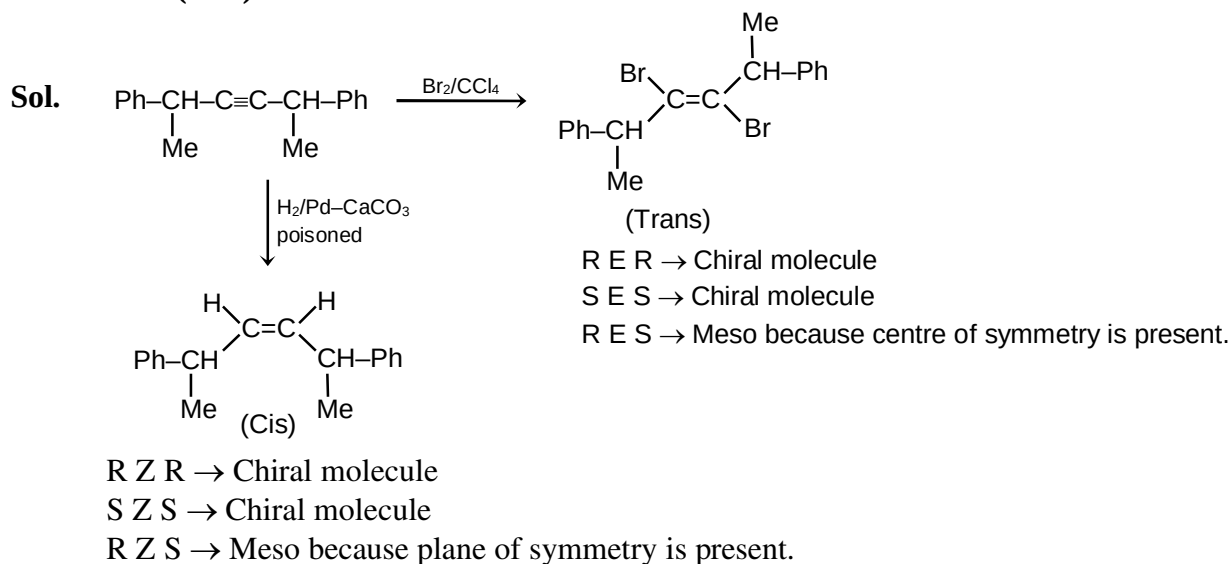
$P_{R_{\text{left}}} = 50 \text{ mm}$

at $t = 5$ hour

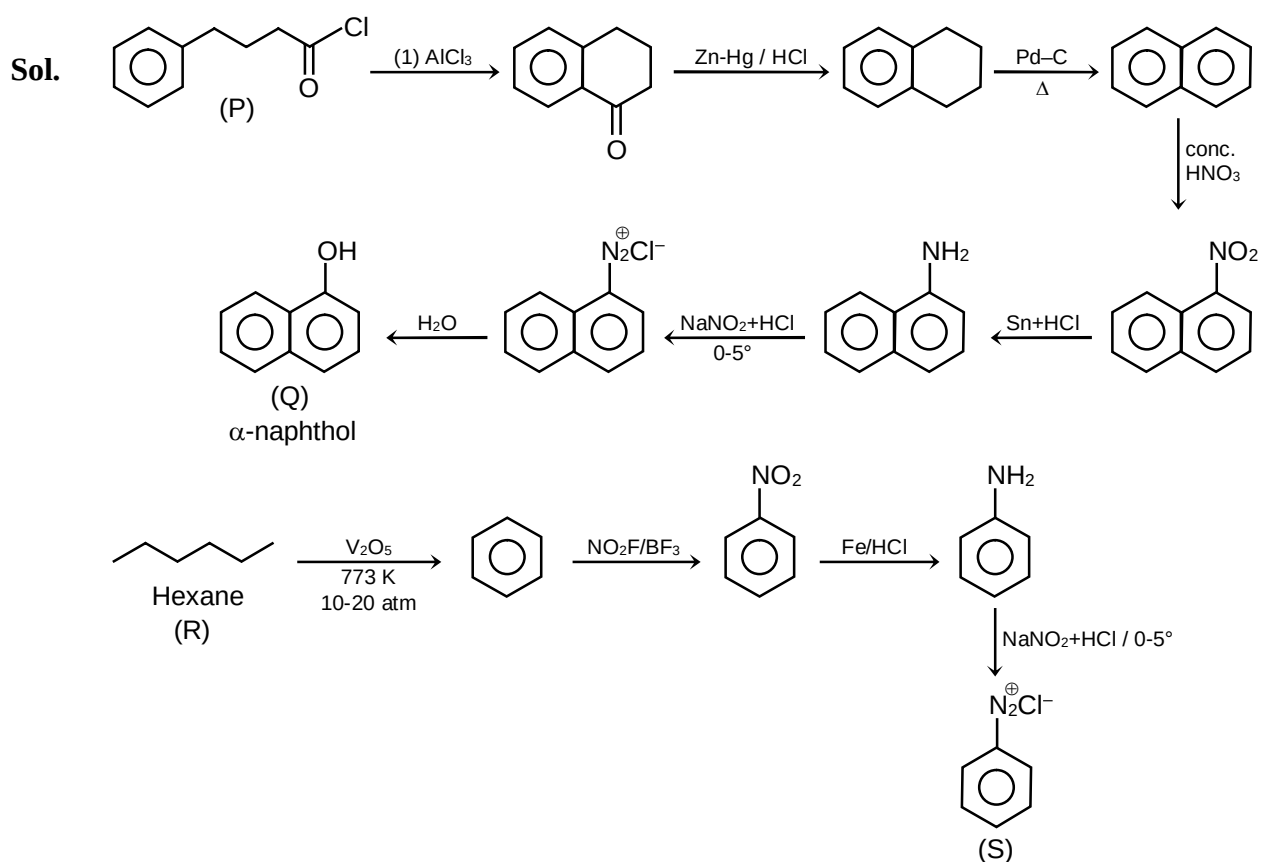
$P_R \text{ reacted} = 750 \text{ mm}$

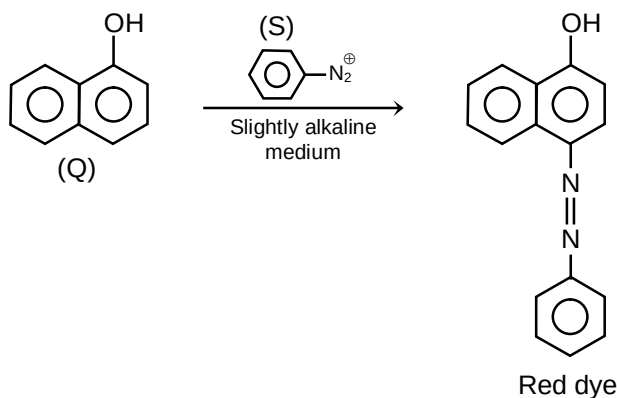
$\therefore P_S \text{ formed} = 375 \text{ mm}$

5. **Ans. (4.00)**



6. **Ans. (19.00)**





PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A,B)

Sol. $(1 - x^3)^n = \sum_{r=0}^n \frac{a_r x^r (1-x)^{3n}}{(1-x)^{2r}}$

$$\Rightarrow \left(\frac{1-x^3}{(1-x)^3} \right)^n = \sum_{r=0}^n a_r \left(\frac{x}{(1-x)^2} \right)^r$$

$$\Rightarrow \left(\frac{1+x+x^2-3x+3x}{(1-x)^2} \right)^n = \sum_{r=0}^n a_r \left\{ \frac{x}{(1-x)^2} \right\}^r$$

$$\Rightarrow \left(1 + \frac{3x}{(1-x)^2} \right)^n = \sum_{r=0}^n a_r \left\{ \frac{x}{(1-x)^2} \right\}^r$$

$$\Rightarrow \sum_{r=0}^n {}^n C_r 3^r \left(\frac{x}{(1-x)^2} \right)^r = \sum_{r=0}^n a_r \left\{ \frac{x}{(1-x)^2} \right\}^r$$

$$\Rightarrow a_r = {}^n C_r 3^r$$

2. Ans. (A,C)

Sol. $A + B^T = B^2$
 $B + A^T = A^2$

$$\Rightarrow (A - A^T) - (B - B^T) = B^2 - A^2$$

$$\Rightarrow B^2 - A^2 \text{ is skew symmetric matrix of odd order}$$

$$\Rightarrow |B^2 - A^2| = 0$$

Now take transpose of equation $B^T = B^2 - A$

$$\Rightarrow B = (B^T)^2 - A^T$$

$$\Rightarrow (B^T)^2 - B = A^T$$

$$\Rightarrow (B^T)^2 - B = A^2 - B$$

$$\Rightarrow (B^T)^2 - B = (B^2 - B^T)^2 - B$$

$$\Rightarrow B^4 - B^2 B^T - B^T B^2 = 0$$

$$\Rightarrow B^2 (B^2 - B^T) = B^T B^2$$

$$\Rightarrow |B|^2 |B^2 - B^T| = |B|^3$$

$$\Rightarrow |B|^2 |A| = |B|^3, \text{ similarly } |A|^2 |B| = |A|^3$$

$$\Rightarrow \text{If } |B| \neq 0 \text{ then } |A| = |B| \text{ and if } |B| = 0 \text{ then } |A| = 0$$

$$\Rightarrow |A| = |B|$$

3. **Ans. (A,B,C,D)**

Sol. $x\vec{a} + y\vec{b} - \vec{c} = \vec{0} \Rightarrow$ If $x + y - 1 = 0$ then C lies on line AB.

If $x > 0, y > 0$ and $x + y - 1 = 0$ then C lies on line segment AB.

If $x < 0, y > 0$ and $x + y - 1 = 0$ then C lies on line but not on line segment AB.

If $x > 0, y < 0$ and $x + y - 1 = 0$ then C lies on line but not on line segment AB.

If $x > 0, y > 0$ and $x + y - 1 < 0$ then C lies inside $\triangle AOB$.

4. **Ans. (A,D)**

Sol. $h'(x) = f(x)$

$h''(x) = f'(x)$

$h(1) = 0, f(1) = f'(1) = h'(1) = h''(1) = 1 = g(1)$

$\Rightarrow f(g(x))g'(x) = 1$

$\Rightarrow f(g(1))g'(1) = 1 \Rightarrow f(1)g'(1) = 1 \Rightarrow g'(1) = 1$

$G(x) = x^2g(x) - xh(g(x))$

$G'(x) = 2xg(x) + x^2g'(x) - h(g(x)) - xh'(g(x))g'(x)$

$= 2xg(x) + x^2g'(x) - h(g(x)) - x^2g'(x) = 2xg(x) - h(g(x))$

$G''(x) = 2g(x) + 2xg'(x) - h'(g(x))g'(x) = 2g(x) + xg'(x)$

5. **Ans. (A,B,C)**

Sol. $\int_0^1 x^{101-n} (A + Bx^{10}) \sqrt{x^{2n} + x^{2n+10}} dx$

$\Rightarrow \int_0^1 (A \cdot x^{(101-n)} + Bx^{(111-n)}) \sqrt{x^{2n} + x^{2n+10}} dx$

$\Rightarrow$ Let $x^{2n} + x^{2n+10} = t$

$\therefore (2nx^{2n-1} + (2n+10)x^{2n+9})dx = dt$

By comparison with $(A \cdot x^{101-n} + Bx^{111-n})$

$\Rightarrow 2n - 1 = 101 - n$

$\Rightarrow 3n = 102$ or $n = 34$

$\therefore A = 68, B = 78$

$\therefore \left(\int_0^2 \sqrt{t} dt \right)^2 = \left(\frac{3}{2} (2)^{3/2} \right)^2 = C$

$\Rightarrow 32 = 9C$

6. **Ans. (A,B,C)**

Sol. The two quadratics are 180° rotations of each other about $(50, 0)$. Since we only dealing with difference of roots, we can translate them to be symmetric about $(0, 0)$.

Now $\gamma = -\beta = 75$ and $\alpha = -\delta$

But $\gamma = 75$ so $\beta = -75$

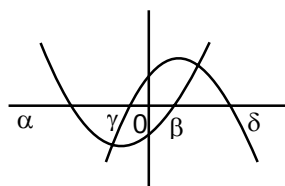
$\therefore p(x) = a(x - 75)(x - \alpha)$

$q(x) = -a(x + 75)(x + \alpha)$

$\therefore p\left(\frac{75+\alpha}{2}\right) = q\left(\frac{75+\alpha}{2}\right)$

$(\alpha - 75)^2 = (\alpha + 225)(3\alpha + 75) \quad (\alpha < 0)$

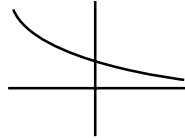
$\therefore |\delta - \alpha| = |2\alpha| = 450 + 300\sqrt{2}$



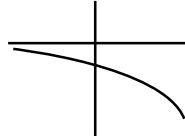
7. **Ans. (A,C,D)**

Sol. Only two cases are possible

Case-I : $f(x) > 0, f'(x) < 0, f''(x) > 0$



Case-II : $f(x) < 0, f'(x) < 0, f''(x) < 0$



Because curve of $y = f(x)$ does not cut the x-axis so curve $y = f^{-1}(x)$ does not cut the y-axis. If $y = f(x)$ cuts the y-axis at $(0, 1)$ means $f(x)$ is always positive. This implies range of $f(f(x))$ cannot lie outside the range of $f(x)$ for $x > 0$

$\Rightarrow f(f(x))$ lies between $(0, 1)$

8. **Ans. (A,B,C)**

Sol. Putting $x = -2$, we obtain (A)

Multiply by x and differentiate w.r.t. x , to obtain

$$A_0 + 2A_1x + 3A_2x^2 + \dots + (n+1)A_nx^n \\ = (x+1) \dots (x+n) + x(x+2) \dots (x+n) + \dots + x(x+1) \dots (x+n-1)$$

Put $x = 1$ to obtain (b)

For (c) replace x by $1/x$ and then multiply by x^n to obtain

$$A_0x^n + A_1x^{n-1} + \dots + A_{n-1}x + A_n = (1+x)(1+2x) \dots (1+nx)$$

Differentiate w.r.t. x to obtain

$$nA_0x^{n-1} + (n-1)A_1x^{n-2} + \dots + A_{n-1} \\ = (1+2x) \dots (1+nx) + 2(1+x)(1+3x) \dots (1+nx) + \dots + n(1+x) \dots (1+n-1)x$$

Put $x = 1$, we get

$$nA_0 + (n-1)A_1 + \dots + A_{n-1} \\ = (3)(4) \dots (n+1) + 2(2)(4) \dots (n+1) + \dots + n(2) \dots (n)$$

$$= (n+1)! \left[\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \dots + \frac{n}{n+1} \right]$$

9. **Ans. (D)**

10. **Ans. (C)**

Sol. (I) & (II)

$$ax^4 + bx^3y + cx^2y^2 + dxy^3 + ay^4 = (ax^2 + \lambda xy - ay^2)(x^2 + \mu xy - y^2)$$

$$\Rightarrow b = \lambda + a\mu \quad \dots (1)$$

$$c = \lambda\mu - 2a, \quad \dots (2)$$

$$d = -a\mu - \lambda \quad \dots (3)$$

$$\Rightarrow b + d = 0 \text{ (from (1) and (3)) and } \lambda\mu = c + 2a \quad \dots (4)$$

Now bisector of $ax^2 + \lambda xy - ay^2 = 0$

$$\text{is } \frac{x^2 - y^2}{a + a} = \frac{xy}{\lambda/2} \Rightarrow \frac{x^2 - y^2}{4xy} = \frac{a}{\lambda}$$

which is same as $x^2 - y^2 + \mu xy = 0$

$$\Rightarrow -\frac{\mu}{4} = \frac{a}{\lambda} \Rightarrow \lambda\mu = -4a \quad \dots\dots\dots (5)$$

from (4) and (5) we get $c + 6a = 0$

Sol. (III) & (IV)

Let roots of $am^4 + dm^3 + cm^2 + bm + a = 0$ are m_1, m_1, m_2, m_2 then $(m_1 m_2)^2 = 1$

$$\Rightarrow m_1 m_2 = 1 \quad \dots\dots\dots (1)$$

$$m_1^2 + m_2^2 + 4m_1 m_2 = \frac{c}{a}$$

$$\Rightarrow m_1^2 + m_2^2 = \frac{c}{a} - 4 \quad \dots\dots\dots (2)$$

$$2(m_1 + m_2) = -\frac{d}{a} \Rightarrow m_1 + m_2 = -\frac{d}{2a} \quad \dots\dots\dots (3)$$

from (1), (2) and (3) we get

$$\frac{d^2}{4a^2} = \frac{c}{a} - 4 + 2$$

$$\Rightarrow \frac{d^2 - 4ac}{4a^2} = -2 \Rightarrow 8a^2 - 4ac + d^2 = 0$$

Answer the question number 11 and 12 by appropriately matching the lists based on the information given in the paragraph

11. Ans. (D)

12. Ans. (C)

Sol. H.P. is $\frac{60}{6}, \frac{60}{5}, \frac{60}{4}, \frac{60}{3}, \frac{60}{2}, \frac{60}{1}$

H.P. is 10, 12, 15, 20, 30, 60

$a = 15, b = 30, c = 60$

SECTION-II

1. Ans. (1.00)

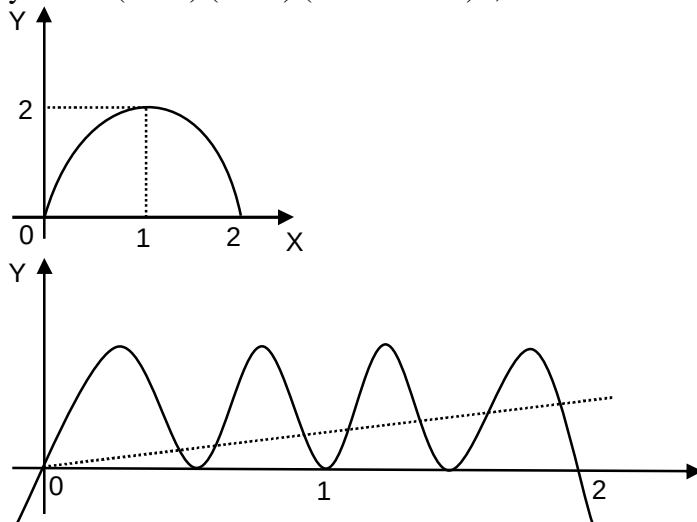
$$\begin{aligned} \text{Sol. } 2 &= (a_{11} + a_{12} + a_{13} + a_{14}) + (a_{41} + a_{42} + a_{43} + a_{44}) + (a_{11} + a_{22} + a_{33} + a_{44}) + (a_{14} + a_{23} + a_{32} + a_{41}) \\ &\quad - (a_{12} + a_{22} + a_{32} + a_{42}) - (a_{13} + a_{23} + a_{33} + a_{43}) \\ &= 2(a_{11} + a_{14} + a_{41} + a_{44}) \end{aligned}$$

2. Ans. (16.00)

Sol. $f(x) = 2x(2 - x)$

$$y = f(f(f(x))) ; 0 \leq x \leq 2$$

$$y = 32x(2 - x)(x - 1)^2(2x^2 - 4x + 1)^2 ; 0 \leq x \leq 2$$



3. **Ans. (16.50)**

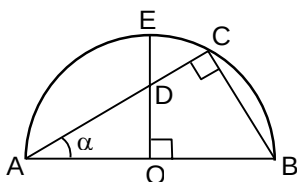
Sol. Let $f(x) = \frac{x^2}{200 + x^3}$

$\therefore f'(x) = 0 \Rightarrow x = (400)^{1/3}$, which lie in (7, 8)

Also $T_7 > T_8$

4. **Ans. (3.00)**

Sol. $OB + CD = OD + BC$



$$1 + 2 \cos \alpha - \frac{1}{\cos \alpha}$$

$$= \tan \alpha + 2 \sin \alpha \Rightarrow \cos \alpha + \cos 2\alpha = \sin \alpha + \sin 2\alpha \left(\cos \frac{\alpha}{2} \neq 0 \right) \Rightarrow \tan \frac{3\alpha}{2} = 1 \Rightarrow \alpha = \frac{\pi}{6}$$

5. **Ans. (6.00)**

Sol. $y = mx + \sqrt{a^2 m^2 + b^2}$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

This tangent also touches the circle $x^2 + y^2 = r^2$

$$\Rightarrow \pm r = \frac{0 - 0 + \sqrt{a^2 m^2 + b^2}}{\sqrt{1 + m^2}}$$

$$m = \sqrt{\frac{r^2 - b^2}{a^2 - r^2}} \quad (\because b < r < a) \quad \dots(1)$$

RS passes through $(ae, 0)$ and parallel to PQ

$$\Rightarrow \text{Equation of RS is } y - 0 = m(x - ae) \Rightarrow mx - y - ame = 0$$

Let T be foot of perpendicular dropped from origin on RS, then

$$RT^2 = OR^2 - OT^2 \Rightarrow RT^2 = r^2 - \frac{a^2 m^2 e^2}{1 + m^2} = b$$

$$\therefore RS = 2b$$

6. **Ans. (3.00)**

Sol. $f(2) = 8 - (r-1)4 - 22 + 4r = -ve$

$$f(-2) = -8 - (r-1)4 + 22 + 4r = +ve$$

$\Rightarrow$ one of the integral root can be, -1 or 0 or 1

$$\text{case-I : } x = -1 \text{ is one root} \Rightarrow r = -\frac{11}{3} \text{ rejected}$$

case-II : $x = 0$ is one root $\Rightarrow r = 0$ but for $r = 0$, $f(x) = x^3 - x^2 - 11x$ which does not have all integral roots.

$\Rightarrow r = 0$ (rejected)

case-III : $x = 1$ is one root $\Rightarrow r = 3$ and at $r = 3$ all roots are integral

$$(\text{equation is } x^3 - 2x^2 - 11x + 12 = 0 \Rightarrow (x-1)(x-4)(x+3) = 0)$$